

Forensic Science

for High Schools

Book 3

Jeannette Jolley
and **John Powrie**



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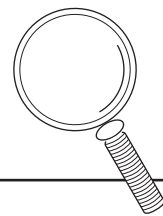
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Design, diagrams and formatting by DiZign Pty Ltd
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Printed in Australia by Ligare Book Printers

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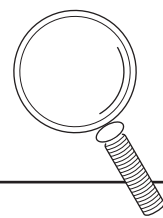
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ISBN 978 1 92253 852 9

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The publisher and the author have made every reasonable effort to ensure that the experiments and activities in this book are safe when conducted as instructed, but assume no responsibility for any damage caused or sustained. Students should carry out all experiments under close supervision.



Forensic Science Book 3 is one in a series of books dealing with the various fields that make up a study in forensics. The series is aimed specifically at the Years 9 and 10 secondary school level, although the material would be suitable for non-tertiary senior courses as well. Book 3 takes a detailed look at two important topics in the field of forensic science. Topic 1 examines the subject of blood and DNA, looking at the way crime scene investigators identify, collect and test samples of biological matter in order to help solve crimes. Topic 2 outlines the role of the forensic scientist at traffic accident scenes, examining the role speed and alcohol consumption play in many serious traffic accidents.

Book 1 introduces forensic science to the student in an interesting and plainly written format. Book 2 looks at Fingerprints and Forgery, and Book 4 examines the importance of Casts, Impressions and Soil. After Book 1, the books may be taught in any order. Depending on individual course structures, certain items, topics or whole books can be omitted so that a forensic unit of any size can be integrated into an existing science course. Each book is approximately 10 weeks' (one term's) work.

Each book in this series contains factual information on several distinct subject areas. The information is interspersed with interesting practical activities and revision questions. The books provide totally prepared units of work, with flexibility to allow teachers to personalise the subject to suit their classes.

The Authors

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Acknowledgments

The authors would like to thank the following people for their support and advice in providing photographs and checking the accuracy of the text:

Dr James Robertson, Director of Forensic Services, Australian Federal Police, ACT

Craig Petterd, Federal Agent, ACT Crime Scenes, Australian Federal Police

Daryl Neit, Accident Investigation Team, Australian Federal Police, ACT

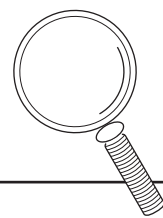
Dr Alastair Ross, Australian National Institute of Forensic Science

Photos

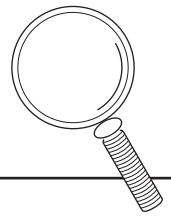
page 19: Photo with kind permission of Professor Dr Alec Jeffreys.

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page 25: Photo courtesy of ACT Traffic Accident Investigation Team, AFP.



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Blood is probably the most important fluid that the forensic biologist has to deal with at a crime scene. It makes up about 9% of a healthy person's body weight, and the average adult has about 5 to 6 litres of blood in their body. When blood is spilt, it goes a long way and is extremely difficult to remove completely. Murderers' attempts to clean up usually fail because blood tends to stick to clothing and the murder weapon, and is splashed around widely in a violent crime.

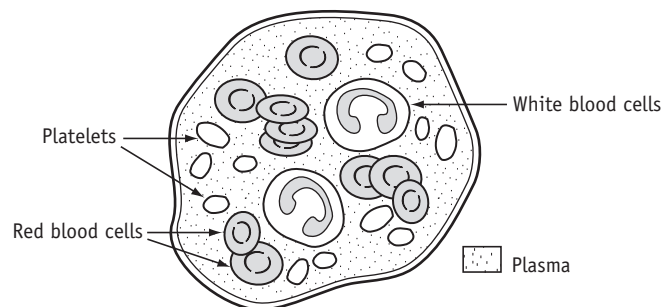
Blood found at crime scenes has trapped many killers who thought that they had removed all incriminating traces. Tests for blood are so sensitive that they will show up the presence of blood even after a surface has been wiped so that no blood is visible to the unaided eye. Blood that is many years old can still be identified as blood and analysed for DNA.

Forensically, blood can yield a great deal of information. The first task for an investigator at a crime scene is to determine whether any suspicious stains are blood by doing a presumptive blood test. If the test is positive, then it must be determined if the blood is animal or human. Once this is established, the stains can be further tested for approximate age, sex, blood group and now DNA.

Understanding how bloodstains and splashes appear under different circumstances can help enormously in reconstructing the sequence of events in a violent crime. Bloody fingerprints, palm or foot prints also tell their own story and may be very helpful in solving the crime.

Composition of blood

Blood is a very complex mixture. It consists of four main components: **red blood cells**, **white blood cells** and **platelets**, all suspended in a pale yellow liquid called **plasma**.



Plasma consists of 90% water and 10% dissolved proteins, hormones salts, digested food materials and waste products.

Red blood cells are a donut shape, and give blood its colour. They carry oxygen to the cells.

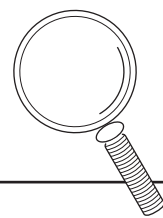
White blood cells are larger and less numerous than the red blood cells. They help destroy dangerous foreign disease-causing organisms.

Platelets are tiny cell fragments which play an important role in blood clotting.



Questions

1. How much blood does the average adult have in their body?
2. What information can blood found at a crime scene give investigators about the victim?
3. What other information can blood splashes give about how a crime was carried out?
4. In your science book, list the four main components of blood and give the main function of each component.



Before the 1900s, it was well known that if blood from one person came in contact with the blood of another person, very often the red blood cells would form clumps and so no longer be able to carry out their role.

In 1901, Austrian biologist, Karl Landsteiner, discovered that some people had certain antigens that could affect blood that did not have that antigen. The different blood types were named **A** (if it had only antigen A), **B** (if it had only antigen B), **AB** (if it had *both* A and B antigens) and **O** (if it had *neither* A nor B antigens). It was found that these blood types contain antibodies that react against the antigens as shown in the table below.

Blood group	Red blood cells contain	Antibody in plasma
A	A antigen	anti-B
B	B antigen	anti-A
AB	A & B antigens	no antibodies
O	no antigens	anti-A & anti-B

Every living person belongs to one of these four blood groups; the group being decided at the moment of conception and inherited from the parents. Hence the blood group is fixed for life and does not change. This ABO system of blood grouping interests forensic scientists because even if dried, the antigens can be detected and identified.

Nowadays the technique of DNA profiling can use blood to link a specific individual to a bloodstain. However, even before this development, a blood sample was considered important evidence if found at a crime scene.

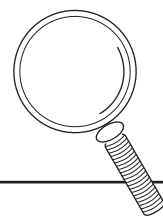
Blood is a good example of a **class characteristic** because it is associated with a group of people all having the same characteristic (ie the same blood group). It is not unique to only one individual, like a fingerprint. Such evidence is still very important during investigations because it eliminates some suspects and so decreases the number of people who need to be examined further.

This type of evidence allows the calculation of the probability that a sample of blood found at a crime scene came from a particular person. To do this, the proportion of the population belonging to that group or class needs to be known.

The table below shows the percentage of each blood type for some different populations.

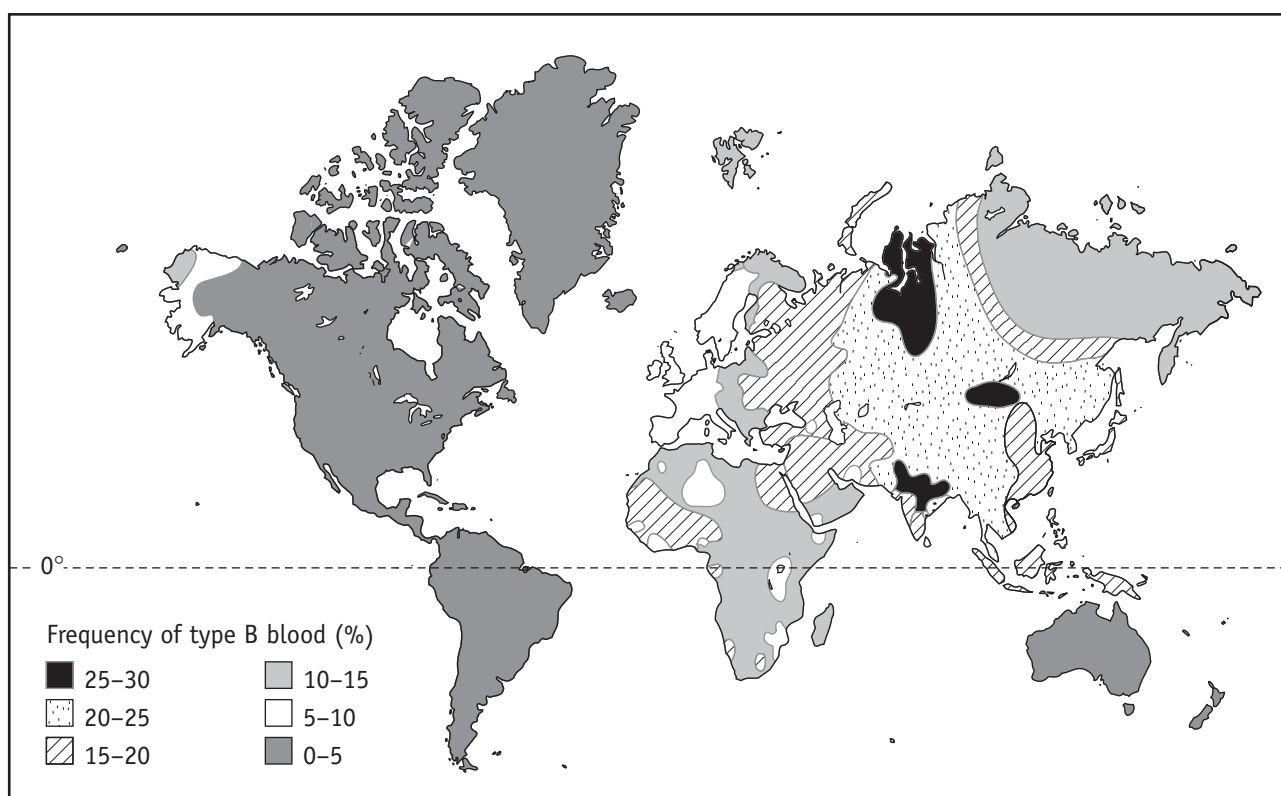
Population	Blood group			
	A	B	AB	O
Australian Aboriginal	66	0	0	34
Australian	39	11	4	46
UK / USA	41	10	4	45
United States Negroes	28	20	5	47
African pygmies	30	29	10	31
African bushmen	34	8	2	56
Pure Peruvian Indian	0	0	0	100
Polynesian	52	0	0	48

For example, if some B type blood was found at a crime scene in Australia, then all the people having a different type of blood (approximately 89% of the population) could be eliminated from further investigation. Hence the probability that a particular person falls into the B group is 11% or approximately 1 in 10. The rarer the blood group the more useful the evidence.



Distribution of blood groups

The table on the previous page shows how some populations have a very different distribution of the four blood groups. Distribution of group B blood is particularly interesting, because some population groups do not have this group at all, while in others it is quite rare, and yet it is quite common in other groups. The map below shows how the frequency of type B blood changes through out the world. Some scientists believe that the changes in the distribution of the various blood groups can be used as evidence to show the way in which ancient populations migrated around the world.



In forensic science the frequency of the different blood groups can be important. If a criminal shed some O type blood at a crime scene it would rule out about 54% of the population as suspects, but could not say which of the remaining 46% committed the crime. However, the most important aspect of blood typing has been to eliminate innocent suspects when their blood type has not matched that left at a crime scene.

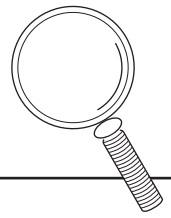
Other blood groups

In 1927, Dr Landsteiner and his team discovered further blood grouping factors that they named M & N. Later, in 1940, they discovered another very important antigen that reacted to the blood of the Indian Rhesus Monkey. This was named the Rh factor and humans can be classified as either 'Rh positive' if they have this antigen, or 'Rh negative' if they do not.

Approximately 82% of Australians are Rh+ and 18% of Australians are Rh-.

The Rh factor allows blood to be sorted into eight main groups: A+, A-, B+, B-, O+, O-, AB+, AB-.

Further research has revealed a large number of proteins and enzymes that are found in plasma. Most are not stable once exposed to the air, but it is now possible to place an individual into one of 288 different blood groupings, which can greatly increase the *probability* of linking a particular individual with a crime.



Secretors

Another discovery that has proven very important to the forensic biologist is that 80% of people are **secretors**. This means that a small number of blood cells are found in their other body fluids such as semen, urine, perspiration and saliva. For these people it is possible to determine their blood group by examining these fluids.

How blood groups are inherited

A person's blood group is inherited from his or her parents. Look at the table below to see how blood groups are passed on.

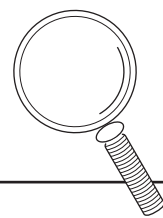
Parent 1	Parent 2	Possible Children's Groups	Impossible Children's Groups
O	O	O	A, B, AB
O	A	O, A	B, AB
O	B	O, B	A, AB
A	A	O, A	B, AB
A	B	AB, A, B, O	All possible
B	B	O, B	A, AB
O	AB	A, B	O, AB
A	AB	AB, A, B	O
B	AB	AB, A, B	O
AB	AB	AB, A, B	O

Research into hereditary factors in blood grouping was often used for the purpose of establishing paternity. It cannot state with certainty that a child does come from specific parents, but it can determine whether a child is NOT the offspring of a given couple.

?

Questions

1. What discovery, made in 1901, meant that less people would die when they were given blood from someone else?
2. What is the most common blood group in the ABO system amongst the Australian population?
3. If some A type blood was found at a crime scene, and one suspect had this same type of blood, what conclusions could you come to regarding the guilt of that suspect?
4. Which blood group of the four ABO types would be the most useful evidence to find at a crime scene? Explain why.
5. Where in the world do you think the B type blood may have originated? Describe how you think it may have spread across the world.
6. Explain what is meant by the term 'secretor'. How this can help in solving a crime?
7. Calculate the proportion of Australians who belong to the following blood groups in the ABO/Rh systems: A + , AB - , B - .
8. A murder victim has AB type blood. A suspect is found, who has a bloodstain of type AB on his shirt. Do you think this is enough evidence to prove the suspect's guilt? What would you think if the victim's blood group and the stain were both O + ?
9. What do you think might happen to a person who has been transfused with blood containing an antibody different to his or her own blood antigen? Explain why this will occur?
10. Find out in what particular situations a doctor needs to know a patient's Rh factor and that of the patient's partner. If a couple's Rh factors are incompatible, what can be done to avoid the death of their children?



If a dark stain is found at a crime scene, it may be blood belonging to either the victim or criminal. Or it could be something else, like furniture polish or another dark stain. Before sending a sample to the lab for testing, the Crime Scene Examiner must first establish that the stain is blood. For this reason, a presumptive blood test is done. This is a test that is quick and simple to carry out at the crime scene, confirming that the stain might be blood and so deserves further investigation.

There are two common ways such a test is done. In many countries a KM (Kastle-Meyer) or LMG (leucomalachite green) test is done. In Australia, forensic investigators use a 'hemastix'. This is simply a plastic strip with a small square of special indicator paper attached to one end. Details of both procedures are described below.



Materials

- phenolphthalein (Kastle-Meyer or KM) solution
- leucomalachite green (LMG) solution
- hydrogen peroxide
- a number of small pieces of filter paper
- pipettes, if solutions are not in dropping bottles

Alternatively, use a number of hemastix.

- some dark brown or maroon stains, including dried animal blood, on a white ceramic tile



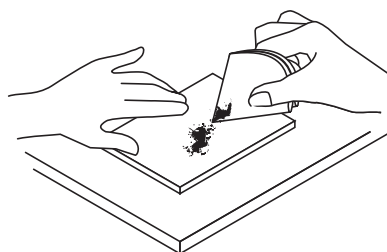
Procedure

1. Students should copy out the results table from the next page into their science books and use it to record observations. The demonstrator should follow either steps 2 to 4 or step 5 below.

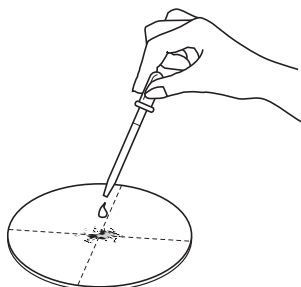
2. Fold a small filter paper into quarters.



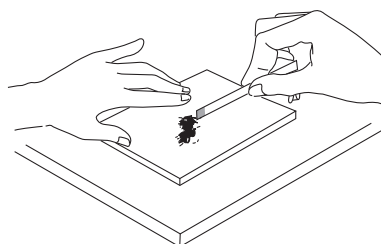
3. Scrape the dried stain with the sharp corner of the filter paper.

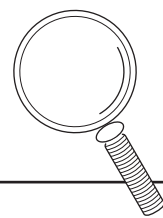


4. Open out the filter paper and add one drop of KM or LMG solution to the stain in the centre. Add a drop of hydrogen peroxide. If there is blood present, you should see an instant colour change.



5. Use distilled water to moisten the indicator square on the end of the 'hemastix'. Gently rub it on the brown stain. If there is blood present, you should see an instant colour change.





Results

Material	Observations when presumptive blood test was carried out
Blood stain	

Human or animal blood?

The test used in the forensic laboratory to determine if blood is of human or animal origin is called the **Precipitin** or **Uhlenhuth Test**, named after the German scientist who developed it. Blood of every animal species contains different proteins. Blood from one species will not accept proteins from another species. This principle is used to test whether bloodstains are human or animal and, if animal, which type.

Serum for the Precipitin test is obtained from rabbits that have produced antibodies to destroy a small quantity of human blood injected into them. A drop of this anti-human serum is added to a sample of the suspect blood, which will precipitate its protein if the protein is of human origin. Most large forensic laboratories hold this anti-sera for many common animals, thus allowing the forensic biologist to confirm or disprove statements made by suspects about where they think certain dark brown stains may have come from.

This test is so sensitive it has been successful in identifying human blood that is fifteen years old and has even been used successfully on tissue samples from ancient Egyptian mummies.

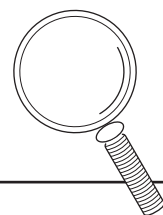


Questions

1. What colour did the blood turn the indicator?
2. What can you conclude regarding the accuracy of these tests to identify blood from the other dark red or brown stains?
3. Discuss how useful you think these tests are when used at a crime scene.
4. Make a list of all the steps a Crime Scene Examiner (CSE) would carry out if a suspicious dark brown stain were found at a crime scene in your local area.
5. If the stain showed positive to blood, what would you expect the CSE to do next?
6. Describe the tests a positive blood stain would undergo in the forensic laboratory.

Information for Teachers: Human blood should never be used in the school laboratory. Animal blood is easily obtained from butchers, or clots can be removed from organs during a class dissection. This blood may be frozen or dried on a ceramic tile for future use in the above demonstration.

In Australian schools we have had difficulty obtaining the correct solutions to give a positive result for blood in both the KM and LMG tests. However, the tests are used extensively in both the UK and USA. In Australia most forensic personnel use Hemastix. These are manufactured by Bayer for urine analysis, and are available from most chemists for about \$25 for 50 strips. Cost is the main reason this activity is recommended as a class demonstration instead of individual practical work by students.



Examination of blood splashes and smears at a crime scene is an essential part of a murder investigation. The position and appearance of blood marks on the body and surroundings will help the investigator to reconstruct the crime.

For example, because blood cannot flow uphill against gravity, it may indicate to the investigator that a body was moved after the victim was killed. French criminologist, Alexander Lacassagne, is credited with linking the shape of blood spots with the position of the victim.

Blood dropped vertically, onto a stationary, flat surface forms a circular mark with a serrated edge. (*see right*)



Drops of blood falling from a moving object hit a flat surface at an angle, leaving a spot like an exclamation mark, with a smaller spot showing the direction of movement. (*see right*) The arrow shows the way the person was moving.



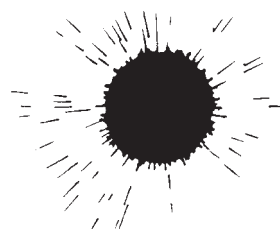
It is sometimes possible to make a rough estimate of the height from which blood fell by inspecting its edges. As the height increases, so does the length of the spines or serrations. (*see right*)



50 cm



100-120 cm



2-3 m

Examination of the shape of obliquely falling blood splashes yields information about the direction and speed of impact. Such evidence helps to determine the positions of the victim and assailant at the time of the assault and may indicate the degree of violence, the type of weapon used and which person was the attacker and who was in a position of self-defence.

The shape of the blood splashes is approximately related to the steepness of the angle of fall. (*see right*) The thickness of the blood, due to the degree of clotting, also has some affect on the shape.

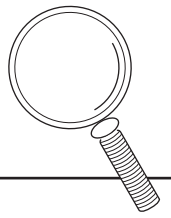


Blood smears tend to start as drops, which become ragged at one edge, indicating direction of travel of the smear. (*see right*) The arrow shows the direction in which the blood was smeared.



When a bloodied surface, like hair, is brushed against a clean surface a distinctive 'swipe' pattern results.

If a clean surface moves against a bloodied one, a slightly different 'wipe' pattern is produced. This commonly occurs when an attempt has been made to clean blood away.

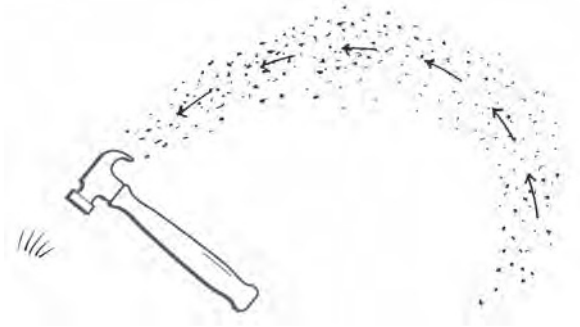


Cast offs

When a bloodied weapon (hammer, axe, fist etc) is moved through the air, droplets of blood may be released behind the moving weapon and form a pattern. If the weapon should stop in mid-air, the droplets may be released and keep going in the same direction the weapon was moving before it was released. These droplets are known as cast offs.

Cast offs may give the following information:

- the direction the assailant's blow came from.
- whether the assailant was right or left handed.
- how many swings were made.
- how the weapon was swung.
- how and where the assailant stood.
- the position of the victim during the attack.



Artery damage and blood loss

An artery is a blood vessel that carries blood under great pressure from the heart to all the parts of the body. Hence blood coming out of a damaged artery will spurt a long way, in big jets, with each heartbeat, resulting in the loss of a lot of blood in a very short time.

Artery splashes are directly affected by the impact and so can be used to locate the origin of that impact.

Estimating the amount of blood lost can help determine the cause of death. If no body is found, it may determine that the victim must have died or that a person having lost so much blood could not have defended himself or herself or caused the assault.



A large puddle of blood may identify the place where a body lay immediately after the crime while the blood still flowed.

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Questions

1. Describe four things that a Crime Scene Examiner may be able to determine from the shape of some blood splashes at a crime scene?
2. Suggest how the splashes below may have occurred.

(a)



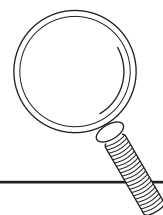
(b)



(c)



3. What is a cast off? Explain how they are formed and what information a Crime Scene Examiner may discover from these spots.
4. If, during a violent attack, a victim's major artery is damaged, discuss the affect of this on the victim, and describe what an investigator might find afterwards at such a crime scene.
5. Describe the steps a junior police officer should follow if suddenly called to a violent crime scene where there is a lot of blood splashed around, and one person is found sobbing over what appears to be a body. Explain the reason for each step that you suggest.



In this activity you are going to investigate how the shape and pattern of blood splashes change under different circumstances and then apply these to cases of criminal investigations.



Materials

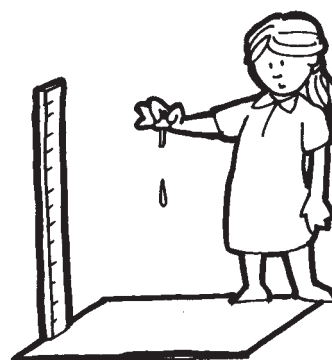
- red (water soluble) dye that has been thickened with gelatine
- newspaper and some butcher's paper for the tests
- metre ruler
- dropping pipette
- some other surfaces that are more porous than the paper (eg cloth) and some less porous (eg white ceramic tile)

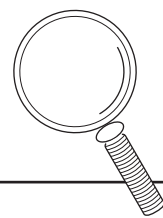
Note: Paint may be given a more blood-like consistency by dissolving gelatine in it.



Procedure

1. Copy the tables shown on the next page into your science book under the heading 'Results'.
2. Cover the floor thoroughly with newspaper then place some butcher's paper over the centre where the 'blood' splashes are going to be dropped.
3. Using a ruler, measure a height of 10 cm above the butcher's paper, and then drop one drop of dye from this height. Label the drop 10 cm, then drop the following drop, next to the first drop, from 30 cm.
4. Repeat this procedure dropping more dye from 60 cm, 1 m and 2 m, or other heights your group chooses to test.
5. Sketch some of the drops in your science books, carefully labelling each with the height. Compare their shapes, and fill in the appropriate table.
6. Now take a board and cover it with butcher's paper then tilt it at about 30° to the horizontal.
7. Drip red dye onto the paper from two different heights (eg 20 cm and 80 cm). Record your results in the third table.
8. Repeat step 7 above with the board tilted at 70° to the horizontal. Try other angles such as 80° or 90° if you have time.
9. Lay some butcher's paper along the floor, and slowly walk along the paper, dropping a drop of dye each time your left foot touches the floor. Measure the distance between the splashes and observe their shape.
10. Repeat step 9, but this time walk quickly.
11. If you have time, drop some dye splashes onto materials that have different surfaces and compare their appearances.





Results

1. DIFFERENT HEIGHTS

Height dropped from	Diameter of solid centre	Other observations
10 cm		
30 cm		
60 cm		
1 m		
2 m		

2. SLOPING SURFACE

Tilt and height	Diameter of centre	Length of splash	Shape of splashes
30°, 20 cm			
30°, 80 cm			
70°, 20 cm			
70°, 80 cm			

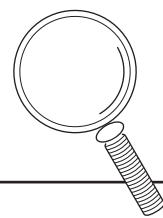
3. MOVEMENT

Speed of walking	Distance between splashes	Shape and size of splashes
slow		
fast		



Questions

1. What change occurred in the appearance of the blood splashes as the height increased? Sketch three spots to show how their shape changed.
2. Use a diagram to show how the shape changes when blood falls onto a sloping surface and describe how the angle of the slope affects the shape.
3. What was the effect of movement on the shape of the spots? Did speed affect their shape? What did you notice about the distance between the spots when the speed was faster? Why would such knowledge be of help to a crime scene investigator?
4. Describe how the thickness of blood can affect the shape of the blood spots. Suggest a way in which you could test your hypothesis.
5. Under what circumstances could blood be thicker than what you would normally expect to flow from a wound?
6. Discuss how knowledge of all these influences on the pattern and shape of blood splashes can contribute to help an investigator solve a crime.



A classic murder case in which blood evidence played a major, if controversial, part occurred in 1954. Dr Sam Sheppard's wife was found dead on a bed in an upstairs room of their home in Ohio, USA. She had been hit by over 30 blows with a blunt object. The room had been ransacked, and was heavily splattered with blood. A trail of stains led down the stairs and onto a terrace.

Dr Sheppard, who had been awakened from sleep on the living-room sofa by his wife's screams, claimed to have been knocked unconscious by an intruder as he rushed up the stairs to help his wife. His behaviour was thought to have been suspicious, and so there was some doubt about his story. This was partly due to his infidelity towards his wife.

The doctor was sent for trial and found guilty of murder. The coroner made much of the bloodstains found on the pillow in the murder room and a bloody imprint, which he suggested, was made by a surgical instrument.

The murder room was full of blood splash evidence which if properly examined would have led to other conclusions. A forensic expert in blood splashes examined the room several months later and came to the following conclusions. As the ceiling showed no traces of blood, the murder weapon must have been used almost horizontally. This was supported by the state of the blood splashes on the walls, some of which appeared to be cast offs flung from the weapon as it was moved backwards and forwards on to the victim's head.

The investigator carried out experiments which suggested that the most likely weapon to have caused the pattern of blood splashes was a heavy torch. He concluded, having noticed blood drops that had been smeared into streaks on the right side of the victim's bed, that the murderer had stood between the twin beds. This suggestion was supported by blood-free areas on two of the walls behind the murderer, which had been protected from flying blood by the assailant's body. A killer standing in that position must have swung the murder weapon with his left hand. Dr Sheppard was neither left-handed nor ambidextrous.

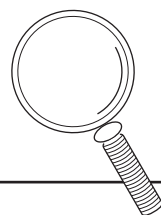
The murderer would have been thoroughly splattered with blood. Yet apart from a bloodstain on the knee of the doctor's trousers, which could have got there when he stood close to the bed to take his wife's pulse, there were no other bloodstains on his clothes.

All these facts came to light during a second trial, and the original investigators admitted neither testing the trail of stains leading from the bedroom to the terrace for human origin nor for blood group. Dr Sheppard was eventually freed.



Questions

1. What forensic evidence was used in the first trial? How did this affect the outcome of the first trial?
2. What forensic evidence was used in the second trial? How did this change the outcome of the second trial?
3. What other forensic evidence that is not mentioned here could possibly have been used?
4. Explain how the evidence given in Q3 could have been useful in this case?
5. What is your conclusion from these events, in terms of forensic science?



DNA is a gigantic molecule found in the nucleus of every cell and in every living organism. It looks something like a ladder with about 3 billion rungs, and is the chemical that **chromosomes** are made of. These are large entwined strands of DNA. They can be divided into smaller sections called **genes**, which are passed down from parent to child. So the DNA from related people is always more similar than DNA from unrelated people.

In the same way, the DNA of closely related animals and plants is also more alike than the DNA of organisms that are only distantly related. It has been found that the DNA of the primates has about 90% in common with the DNA of humans.

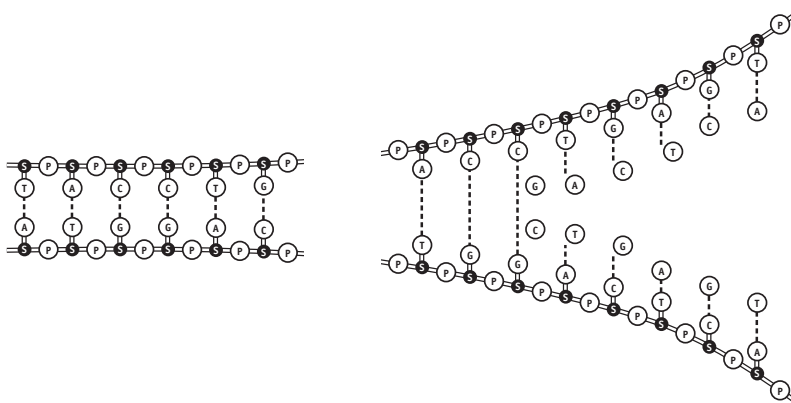
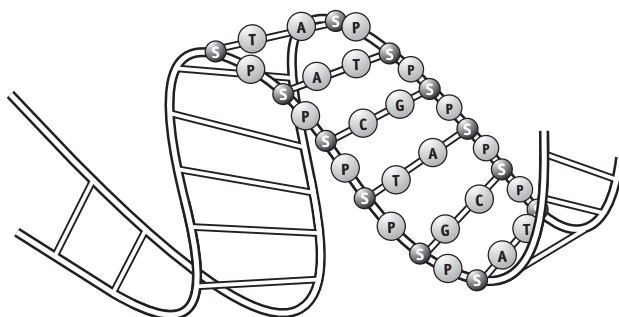
The only people who share exactly the same DNA are identical twins, because they started out as a single fertilised egg, that in its early stages split apart, and then grew into two separate babies who are genetically identical.

The legs of the 'ladder' consist of alternating molecules called **sugars** and **phosphates** (marked as 's' and 'p' in the diagram above). The ladder twists itself into a 'double helix' to save room as it fits into the nucleus. The rungs of the DNA ladder are made up of pairs of molecules called **bases**. There are only four different bases, and they always pair up in exactly the same way:

adenine (A) always pairs with thymine (T), and cytosine (C) always attaches to guanine (G).

The DNA ladder has also been compared to a zip fastener with the four bases forming the teeth. The difference is that in order to join the two strands of DNA, A must pair with T, and G must pair with C. The order in which these bases appear is the **genetic code** that determines all inherited characteristics (*see below*).

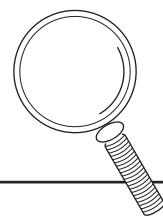
This DNA, or genetic code, is unique to every person and is a combination of which 50% is inherited from the father and 50% from the mother. When a cell divides in two, it opens like a zipper (*see far right*), and the DNA copies this order of bases exactly, so the DNA in the new cell is identical to the DNA in the original cell.



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Questions

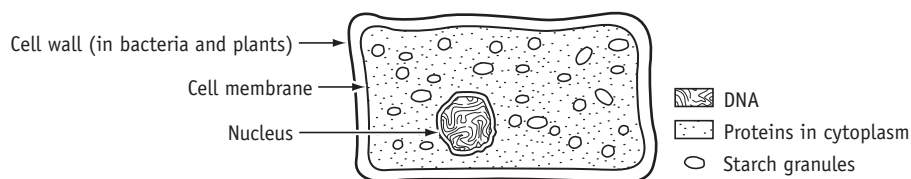
1. Where in our bodies is DNA found? Explain the purpose of this molecule.
2. Describe how DNA is able to make an exact copy of itself when a cell splits into two new cells.
3. DNA is not as unique to all individuals as fingerprints are. Explain the difference and discuss which type of evidence you think is better and why.
4. What is meant by the term 'genetic code'? Explain how this chemical DNA is able to instruct the body to produce red hair, blue eyes or a certain nose shape.



DNA can be extracted from a number of organisms by the following steps:

1. breaking down the cell and nuclear membranes (called 'lysing' the cells with a detergent)
2. breaking down and removing cellular proteins by dissolving them
3. precipitating out the DNA (done with ethanol).

Dried chickpeas are rich in DNA as well as proteins and starch, so these are the components that are removed in this activity. A typical pea cell can be represented by the following diagram.



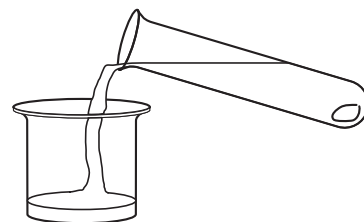
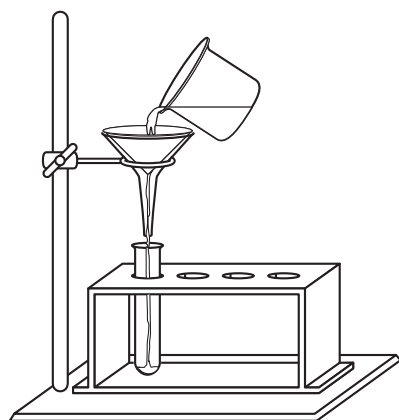
Note: **Solution A** is a buffering and salt solution, **Solution B** is a membrane lysing solution, **Solution C** is a buffering, lysing and salt stabilising solution, and **Solution D** is a nuclear membrane lysing solution. The 'soup' is made before the lesson. See notes at end of next page for details.

Materials

- 10 mL of dried pea 'soup' in small beaker
- Solution C
- Solution D
- Ice cold ethanol
- Cotton gauze
- Centrifuge
- 2 plastic centrifuge test tubes
- 1 small glass test tube
- 3 plastic pipettes
- Glass rod with hook at one end

Procedure

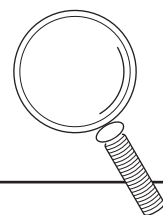
1. Filter 10 mL of the pea 'soup' through gauze into your first plastic test tube.
2. Centrifuge at 3000 rpm until a well-packed pellet is formed. The liquid above the solid pellet contains the unwanted proteins and salts.
3. Pour off this liquid (which is called the supernatant) and **keep the pellet**, which consists of starch and cell nuclei containing DNA.



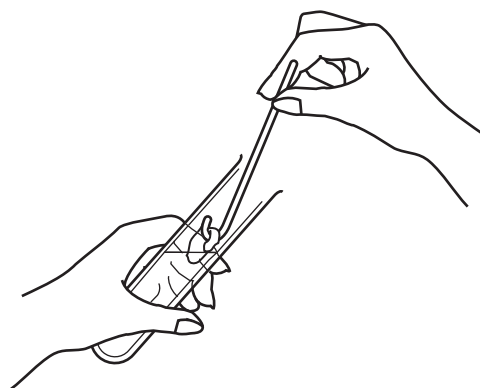
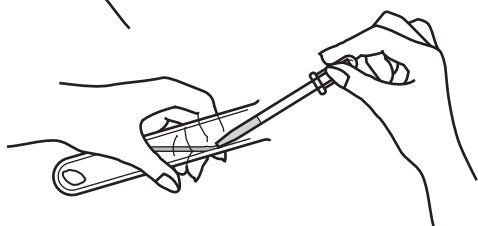
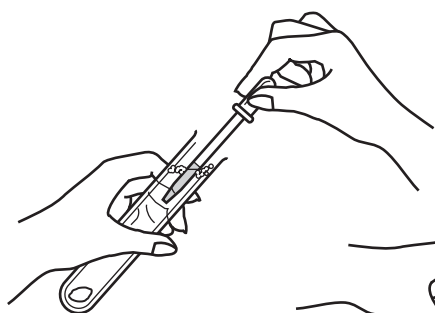
4. Add 10 mL of **solution C** to the pellet left in the test tube. Stir gently with the pipette until the pellet is completely broken up.
5. Centrifuge again at 3500 rpm for 4 minutes until another well-packed pellet is formed and again discard the liquid at the top, which contains more protein.

Activity: Extracting DNA from peas

1 B L O O D A N D D N A



6. Add 4 mL of **solution D** and stir gently with the pipette until all lumps are removed.
7. Centrifuge once more at 3000 rpm to form a well-packed pellet. **KEEP THE LIQUID.** The pellet is mainly starch but the liquid on top (the supernatant) will contain your DNA.
8. Being careful not to disturb the pellet, use a plastic pipette to extract about 2 mL from the **middle** of the supernatant and transfer it into a clean glass test tube. Leave behind the foam on top of the liquid.
9. Tilt the glass test tube and using a clean pipette, slowly add 3 mL of icy cold ethanol down the side so that it forms a layer on top of the DNA solution rather than mixing with it.
10. Now use the glass hook to gently stir the layer where the ethanol touches the DNA solution and observe the formation of the long strands of DNA. If you can scoop the DNA on to the end of the glass rod, you should be able to lift it out of the test tube.



Note:

DNA is fairly transparent. If your DNA appears whitish it is due to contamination with protein that was not completely removed in the earlier steps.

Steps 1 to 5 are to remove protein, and steps 6 to 8 to remove the DNA from the starch.

X

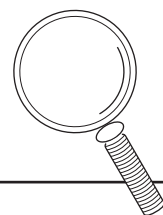
Exercise

1. Write out a description of the **Method** that you used to extract your DNA.
2. Under the heading '**Results**', describe the appearance and quantity of your final product. Compare your product with other groups and describe how successful your group was at extracting DNA from dried peas.
3. If the quantity or quality of DNA that your group extracted was less than others, discuss some reasons why this might have been. Suggest where you might make improvements if you had time to repeat this activity.

Notes for teacher or lab assistant:

- Solution A is 10 mM MgCl_2 , 10 mM Tris buffer pH 7.5 and 0.2 M NaCl.
- Solution B is Triton X-100 (20% in distilled water).
- Solution C is 0.5% Triton X 100, 10 mM Tris, 10 mM MgCl_2 pH 7.5.
- Solution D is 1% SDS in distilled water.

The 'soup' is made before the lesson by grinding 1 g dried chickpeas with 10 mL solution A, then stirring for one hour with 0.25 mL solution B. This is sufficient material for one group, increase proportionally for more groups. This 'recipe' was provided by the CSIRO 'Green Machine'.



This is probably the most important discovery for use in forensic science since the development of fingerprinting over a hundred years ago. While blood grouping can positively exclude a suspect, it can only give a probability as to guilt. DNA profiling can actually prove a person's identity, except in the case of identical twins. No wonder it has been said to have revolutionised crime solving.

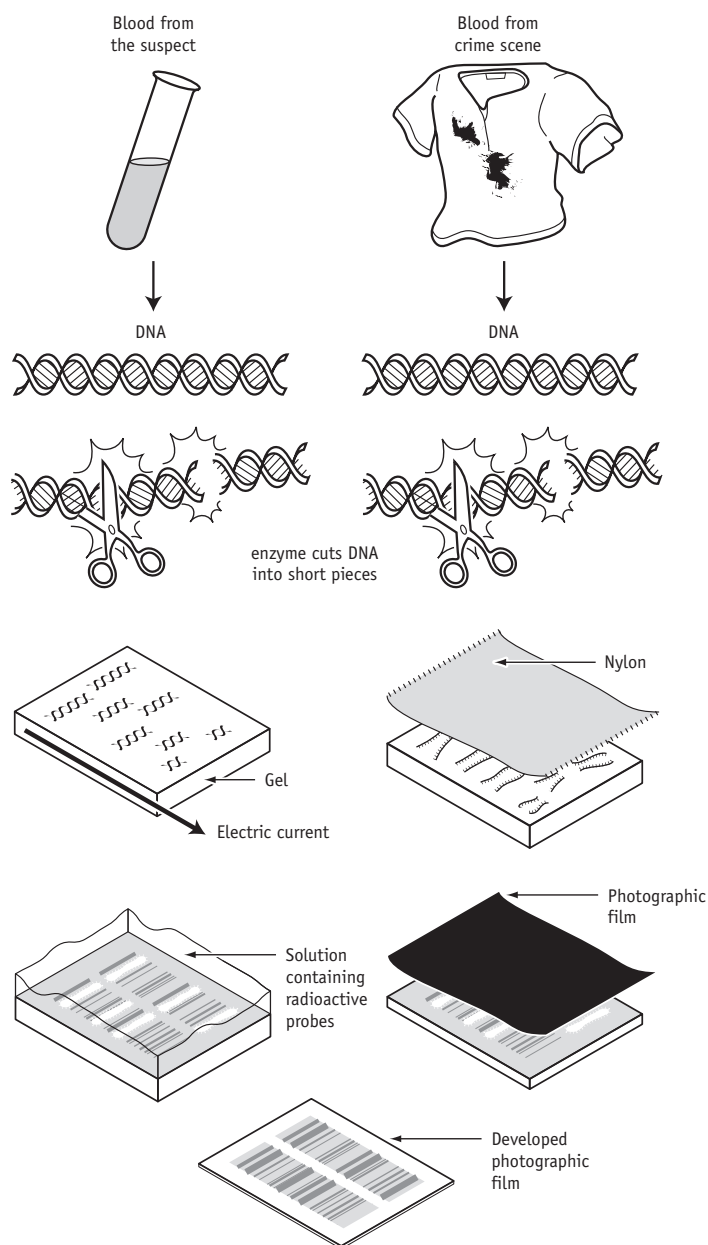
It was in September 1984 at Leicester University, England, that Professor Alec Jeffreys stumbled across his exciting discovery. He was working on parts of DNA where there was a sequence of bases that varied greatly in unrelated individuals. He and his team found a way in which these sequences could be made visible as X-ray lines that looked a little like a bar code. These 'bar codes' give a unique pattern for every person, hence the original term of 'genetic fingerprinting'.

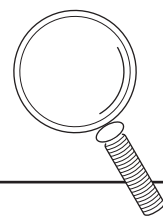


Professor Sir Alec Jeffreys

How a DNA profile was produced

1. A blood or hair sample is collected from a suspect and a garment found at a crime scene.
2. The DNA is extracted in the laboratory, then 'cut' into small pieces using a scissors-like chemical called an enzyme. The lengths of these fragments differ greatly from person to person.
3. Next these pieces are placed in a gel through which an electric current is passed. The current carries the fragments along through the gel. The shorter pieces move faster, so they end up further along than the longer pieces.
4. A chemical is added that causes the two strands of DNA to unlink like a zipper being opened. The sorted DNA fragments are now transferred to a sheet of nylon.
5. Radioactive strands of DNA (called probes) are added, and they link up with some of the fragments. The rest are rinsed away.
6. When the nylon sheet is placed under some photographic film, the radioactive probes expose it, and when developed, they leave a pattern that resembles a bar code.
7. This pattern of dark bands from the suspect's blood is compared to the pattern made by the DNA from the crime scene. If they match, the two samples are from the same person.





Modern developments in profiling

Although the original method was very discriminating, and the odds of a random match were one in several billion, it was very time consuming and hence expensive. In the early days of DNA analysis it could take up to six weeks for a profile to be produced. Also there had to be a sufficient quantity of the biological material and it had to be in good condition. Unfortunately, in forensic situations this is often not the case, because DNA is easily damaged by heat or bacteria. Very often evidence cannot be collected while it is still fresh.

Modern techniques now include an amplification (copying) process where very small amounts of DNA are duplicated over and over until enough has been made to produce a profile. In fact, sufficient DNA can be extracted from only a few hair roots, saliva left on a drink bottle, licked stamp or in chewing gum that has been masticated in the mouth. Even skin cells left on clothes that have rubbed against the body may be used.



A lab technician swabbing for DNA

The downside of this extreme sensitivity is that it is much easier for a sample of biological material to become contaminated with DNA from the person collecting it, or any other innocent person who has come in contact with the crime scene. Hence great care must be taken by anyone entering such an area or carrying out the analysis.

The parts of our DNA that are chosen for analysis are small sections where for some unknown reason a group of bases repeat themselves, like a stutter. These parts do not appear to have anything to do with our inherited characteristics, so it is these sections of DNA that are being amplified and analysed to give a person's profile.

For example, if a sequence of bases is saying **G C T T A C T G T G A T C**, the bases appear to occur completely at random. However, unexpectedly a couple of bases will repeat like this:

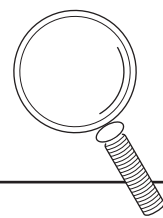
G C T T A C T A C T A C T A C T A C T G T G A T C

It has been found that the number of repeats varies for different people. So Person A might have five repeats as shown above, but Person B has only four repeats, while Person C has twelve repeats of the same short sequence. Now if several of such stutter sites are looked at, the chance of two people having exactly the same number of repeats of the same sections becomes very unlikely.

Most developed countries now use a system called **Second Generation Multiplex (SGM)** which copies ten such sections of the DNA as well as incorporating a sex marker. Using these eleven parts the technique is extremely sensitive and has a discrimination order of 1 in 1000 million. It is a record of the numbers of repeats of these sections that are now held on a country's National Database.

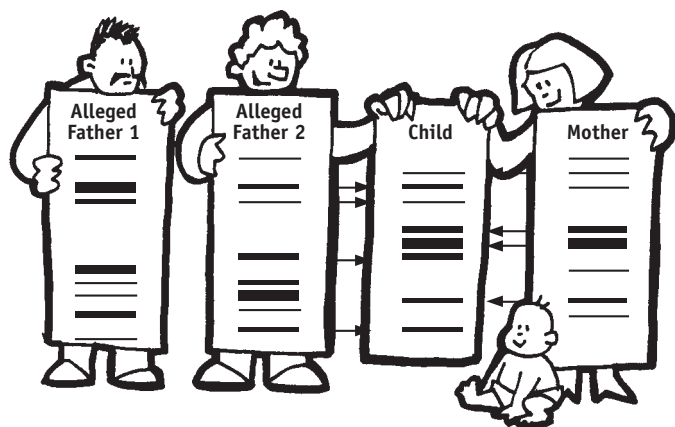
Australia has databases in each state and territory, and most Western countries have similar databases. They contain records of all profiles collected from unsolved crimes and from convicted offenders. Samples from offenders are taken either from blood, mouth swabs (buccal cells), or head hair. Each time a new profile is added, it is searched against all other profiles already in the database. It has been found that an amazing number of previously unsolved crimes have now been solved using these databases.

Analysis now takes only 48 hours and is getting quicker. The cost has also come down so that the technique is being used routinely for non-serious crime like house burglaries and car theft. DNA profiling is now helping to solve many crimes that were not able to be solved previously.



How DNA profiles are matched

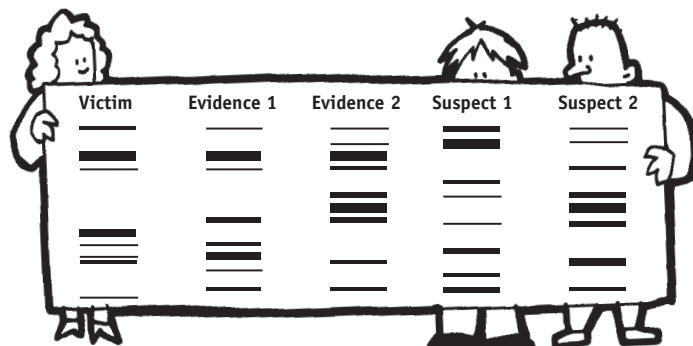
DNA profiling, as well as being extremely important in trapping killers and rapists, is also used routinely to establish the identity of a child's father in paternity disputes, or in proving family links in immigration applications.



A child inherits half its chromosomes, and hence its DNA, from each of its parents. In cases of disputed paternity, DNA profiles are obtained from the mother, the child and both men claiming to be the father. Each band on the child's profile must match one on either the mother's or father's profile as shown left. Hence it is possible to identify the real father with certainty.

The child's real father above is father 2, whose profile has about half the bars in common with the child. The other bars on the child's profile come from the mother.

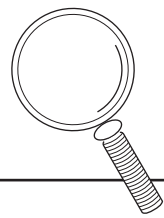
In the diagram on the right, the DNA profiles from the victim and that found at the crime scene (Evidence 1 and 2) closely match the profile from Suspect 2, hence implicating him. Suspect 1's profile has no bars in common with the evidence, so he is most likely innocent of the crime.



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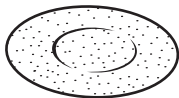
Questions

1. Who is the person credited with the discovery of DNA profiling? When and where was this discovery made?
2. Discuss three disadvantages of the original profiling technique for use in forensic science.
3. Describe the modern developments that have been made to overcome these problems.
4. How has the time taken to produce a DNA profile changed since it was first developed?
5. Describe what is stored on a country's National DNA Database, and describe how it is used to solve crimes.
6. Explain how the setting up of this database has affected the number of crimes being solved.
7. Describe some other uses of DNA profiling besides its use in Forensic Science.
8. Explain how blood found on some window glass at a crime scene can be used to discover who broke the window and climbed in to burgle the house.
9. List the evidence from which DNA may be collected and try to put it in order, from what you think may be the most useful to a forensic scientist, to least useful. Explain your answer.



1. The diagrams represent two types of blood cells. Use them to answer the questions below.

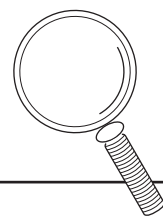
A.



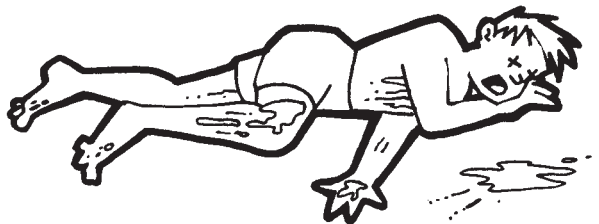
B.



- (a) From their shape, identify which is a red blood cell and which is a white blood cell.
 - (b) Which one is more common in a sample of blood?
 - (c) Which one is used to identify a person's blood group?
2. Describe the two main purposes of blood and explain why this liquid is so important.
3. Blood is said to have 'class characteristics'.
- (a) What does this mean?
 - (b) What other evidence could be described as having class characteristics?
 - (c) Can an individual be identified by his or her blood group? Explain why or why not.
4. What are the four main blood groups in the ABO system?
5. When were these blood groups discovered, where and by whom?
6. Explain how each group differs from the others?
7. There are eight different ABO/Rh blood groups possible.
- (a) Which of these groups is the most common in Australia? What proportion is this?
 - (b) Which of these groups is the least common in Australia? What proportion is this?
 - (c) Explain how your answer would be different if it referred only to native Aboriginal Australians?
8. When does a person's blood group become established, and does it ever change?
9. Which of the four ABO groups can:
- (a) be used to transfuse any other group? Explain why.
 - (b) receive blood from any other group? Explain why.
10. What other blood groups besides the ABO and Rh systems have been discovered?
11. How many different blood groups can a biologist now use to help identify a criminal?
12. If both parents have O type blood, what blood group(s) are possible in their children? Explain why?
13. If one parent has A type blood, and the other has type B, is it possible that both their two children could have O type blood? Explain your answer.
14. A suspect in a murder case is found to have blood group O+ which matches the blood group found at the crime scene.
- (a) Does this mean that the suspect was probably there? Discuss.
 - (b) What if the suspect's blood and that at the murder scene was type AB-? Does that make a difference to the possibility?
15. What are two common chemicals used to test if a dark brown stain is blood? Describe the colour changes that would be found in each case if the stain is blood.
16. Discuss what else, besides blood, may give a positive test to the chemicals described in Q15 above. Explain why these tests are still used at a crime scene when there are some substances other than blood that will give a positive result.
17. What are four things a forensic investigator could conclude from the shape of one or two blood splashes?



18. What important information could an investigator get from a crime scene where a body is found lying down covered in bloodstains as illustrated right?



19. Suggest how the stains below could have occurred.

A.

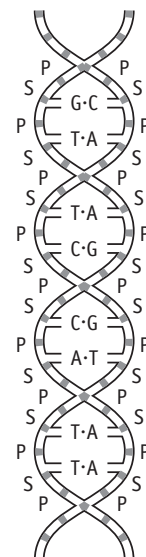


B.



20. Look at the diagram on the right.

- What is it a diagram of?
- What do the letters S and P represent?
- What are A, C, G and T?



21. Name five sources of body tissue a biologist could use to make up a DNA profile?
22. DNA profiling is said to have revolutionised forensic science. Explain why someone might say this.
23. A DNA profile is not unique to every individual. Explain who are the exceptions and why.
24. Who is the real biological father of the child in the profile below? Explain how you came to this conclusion.

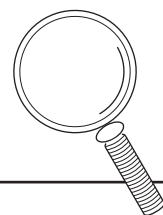
Mother	Child	Father 1	Father 2
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—
—	—	—	—

25. Which suspect appears to have left some body cells at the crime scene? Do you think any of the victim's DNA was left there as well? Explain your answer.

Victim	Crime Scene	Suspect 1	Suspect 2	Suspect 3
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—
—	—	—	—	—

Blood and DNA wordsearch puzzle

1 B L O O D A N D D N A



All the words listed below are hidden in the grid. Use pencils to find the hidden letters, then the remaining 27 letters will spell out a sentence. Write out this sentence in the space below.

A	L	A	C	A	S	S	A	G	N	E	D	H	T	U	H	N	E	L	H	U
D	A	N	L	A	U	D	I	V	I	D	N	I	A	P	U	N	A	M	U	H
E	N	R	O	T	C	A	F	H	R	M	I	A	L	C	I	D	I	U	L	F
N	D	E	O	X	Y	R	I	B	O	N	U	C	L	E	I	C	A	C	I	D
I	S	N	I	K	S	D	M	T	H	E	R	E	L	O	S	U	C	O	L	F
N	T	L	A	E	I	U	O	E	M	L	U	A	V	I	L	A	S	N	S	T
E	E	I	M	P	R	O	R	O	V	S	H	T	W	I	N	S	O	T	P	S
H	I	E	N	D	R	I	S	I	L	T	A	E	W	S	O	P	S	P	L	I
E	N	G	E	R	T	O	C	C	H	B	G	I	S	U	A	E	R	A	A	G
N	E	R	I	E	M	T	F	P	A	S	E	C	R	E	T	O	R	S	S	O
I	R	A	D	O	I	A	L	I	N	L	N	C	W	E	T	I	R	T	H	L
N	H	W	R	M	D	O	T	O	L	C	E	I	S	E	P	A	C	A	E	O
A	O	H	E	C	N	E	D	I	V	E	S	S	I	S	P	V	E	I	S	I
U	C	D	J	E	F	F	R	E	Y	S	S	N	N	E	G	I	T	N	A	B
G	O	R	H	A	M	S	A	L	P	N	O	I	T	A	N	I	M	A	X	E
C	Y	P	L	A	T	E	L	E	T	S	L	L	E	C	A	S	T	O	F	F

ADENINE
ANTIGEN
BIOLOGIST
BLOOD
CAST OFF
CELLS
CHROMOSOME
CLAIM
CLOT
CODE
DEOXYRIBONUCLEICACID
DNA PROFILES
EVIDENCE
EXAMINATION
FLUID
GENES
GUANINE

HAIR ROOT
HELIX
HUMAN
INDIVIDUAL
INHERITED
JEFFREYS
LACASSAGNE
LANDSTEINER
LOCUS
MURDER
NUCLEUS
PHENOLPHTHALEIN
PLASMA
PLATELETS
PROTEIN
RAPE

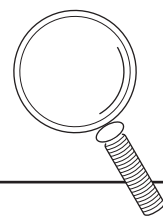
SALIVA
SCALES
SECRETOR
SEMEN
SKIN
SPIRAL
SPLASHES
SOURCE
STAIN
SWEAT
TESTS
TWINS
UHLENHUTH
VICTIM
WEAPON

RH FACTOR (blood antigen that reacts to blood from the Rhesus monkey)



Exercise

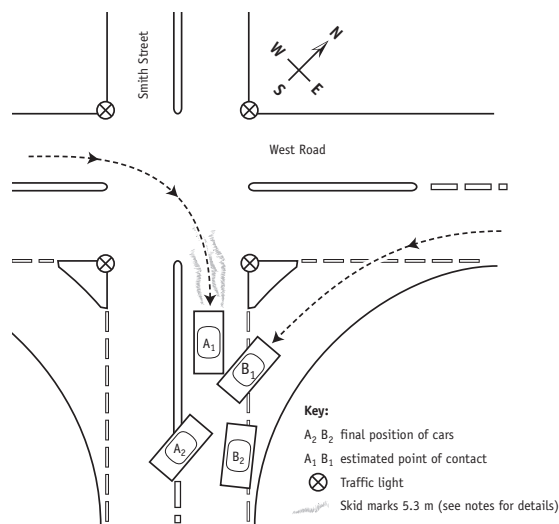
1. Select 15 words from those listed above and write a single sentence clue for each that could be asked in a crossword. (One example 'RH FACTOR', has been done for you)
2. Draw up a crossword grid and clues for these 15 words, all interlocked by at least one letter.



Although the forensic laboratory is not often involved in the analysis of traffic accidents, traffic related crime still takes up a large proportion of police time and resources. The work done by the Accident Investigation Team is very similar to the work done by Crime Scene Examiners.

When an accident occurs, after any injured people have been taken care of, evidence has to be gathered so that the cause of the accident can be determined. Only then can it be decided if criminal charges need to be laid. If the accident has caused any fatalities, then the investigation becomes a coronial inquiry. Police records and statistics show that the two main causes of road accidents are speed and alcohol.

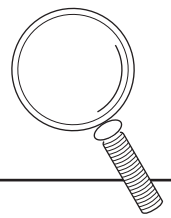
After an accident, a number of photographs will be taken and maps drawn to record the final resting place of all the vehicles involved. Any eyewitnesses to the incident will be interviewed, and the weather and road conditions recorded.



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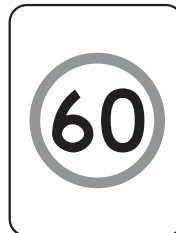
Questions

1. Describe what you see is the role of the Accident Investigation Team after a traffic accident.
2. What other tasks besides the ones you have described in the previous question would need to be carried out after a serious accident? Explain who you think would do each task and why.
3. Look at the sketch of a road accident above. If it were fatal, or resulted in serious injury, the police would have to find out the most likely cause of the accident.
 - (a) Make a list of at least eight measurements the investigators would take to help them reconstruct the incident. Explain the information these measurements would give.
 - (b) What tests do you think would need to be carried out on the drivers of the cars involved? Explain why these tests are done.
 - (c) Describe any tests you think the cars should undergo and explain why.
 - (d) List some possible questions you think an eyewitness to the accident should be asked? What things do you think might determine how reliable such a person's answers would be?
4. Make a list of ten things that may contribute to the cause of a traffic accident. Next to each write who you think is to blame, eg *driver*; *victim*; *car manufacturer*; *road repairer* or *no blame*.
5. When you have finished your list, compare it with one written by a classmate. See if you have some of the same causes and agree as to who is most to blame. Add to your list if you think it appropriate to do so.
6. Do you think that it is possible for an accident to occur in which there was no-one at fault? If so, try to describe one such occurrence and explain what could be done to reduce the chance of it happening.



Being allowed to drive a vehicle on a public road is a privilege and not a right. Today a person wishing to take a vehicle out on the road must have a current driver's licence. To obtain such a licence they have to pass a number of tests to show that they are competent at handling the vehicle, and know the rules that govern the behaviour of all persons on the road.

Once a person has qualified and obtained their licence they are expected to abide by the rules. They should drive within the speed limit set for the piece of road that they are driving on, and be alert, courteous and considerate of other road users. They should not be under the influence of alcohol or other drugs, which will impair their judgement. If a driver breaks these rules and is caught, then punishment usually follows.



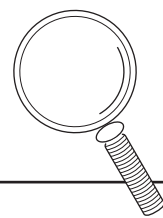
?

Questions

1. Explain why you think that we have road rules?
2. Write a short paragraph describing what you think our roads would be like if there were no road rules.
3. Find out what the range of punishments are for the following misdemeanours on the road:

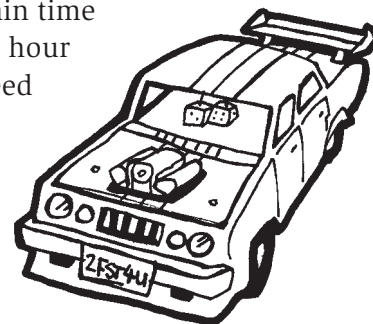
(a) driving 10 km over the speed limit	(e) driving while unlicensed
(b) driving 20 km over the speed limit	(f) not wearing a seat belt
(c) driving under the influence of drugs	(g) not wearing a helmet while riding a bicycle or motorbike
(d) driving while your blood alcohol content is over 0.05 %	
4. Look at the drawing on the right of an accident that occurred at an intersection.
 - (a) Which car would you say was at fault? Why do you believe this?
 - (b) Do you think that the other driver may have contributed in any way? If so, in what way?
5. In any intersection accident, insurance companies now attribute some blame, and hence loss of no-claim-bonus, to both parties.
 - (a) Why do you think they do this?
 - (b) Why do you think that this sharing of blame does not happen in rear-end collisions as drawn on the right where the truck has run into the rear of a car. Only the truck driver loses his no-claim-bonus and the car driver is considered to be without blame.
6. Several days after the Accident Investigation Team measured the length of skid marks at the scene of a collision, the police charged the driver of the car involved with dangerous driving because he had exceeded the speed limit. Explain how they could have known this when there was no eyewitness to the accident.





Speed or **velocity** is a measure of how far an object moves in a certain time period. The most commonly used units of speed are kilometres per hour (km/h), or metres per second (m/s). A speed of 35 km/h is the speed at which a vehicle would take one hour to cover 35 kilometres. A speed of 10 m/s is the speed at which 10 metres is covered in every second.

A speed given in m/sec can be converted to km/h by multiplying by 3.6. Similarly, km/h can be converted to m/s by dividing by 3.6.



Calculating speed

Speed can be calculated using the formula:

$$\text{average speed} = \frac{\text{total distance moved}}{\text{time taken}}$$

This can be written in symbols:

$$v = \frac{d}{t}$$

v = speed or velocity

Example: If you want to calculate the average speed of a car that travels 40 kilometres in half an hour, you divide the distance (40 km) by the time (0.5 hours).

$$40 \div 0.5 = 80 \text{ km/h}$$

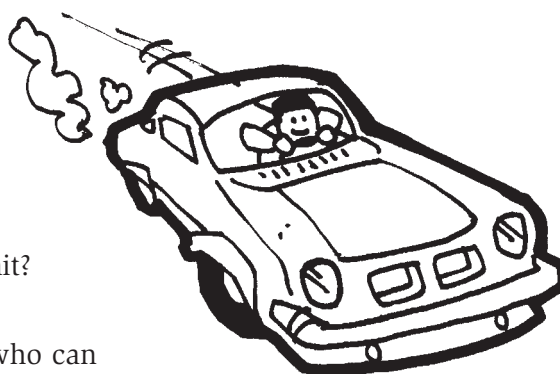
To convert to m/s:

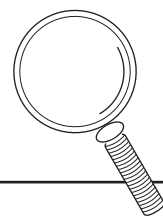
$$80 \text{ km/h} \div 3.6 = 22.2 \text{ m/s}$$

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Questions

1. What is the average speed of my car if I travel 330 kilometres from my house in Canberra to Sydney in 3 hours?
2. What average speed in km/h was a motorcycle doing if it travelled 17 kilometres in 15 minutes?
3. A police motorcycle followed a vehicle for 5 minutes and found that it had covered $3\frac{1}{2}$ kilometres in that time. If the speed limit in this area was 60 km/h, should the driver of the vehicle be charged with going over the speed limit? Show why or why not.
4. What is the average speed in m/s, of a sprinter who can run 100 metres in 10 seconds? What is this in km/h?
5. Some swimmers can swim 1500 metres in 15 mins. What is their average speed in m/s and in km/h?
6. A family going on holidays left home at 9 am. They arrived at their destination 300 kilometres away at 11.30 am. If the speed limit on the highway was 110 km/h, did they exceed the speed limit? If so, by how much? How much longer would the journey have taken if they had kept to the correct speed limit?

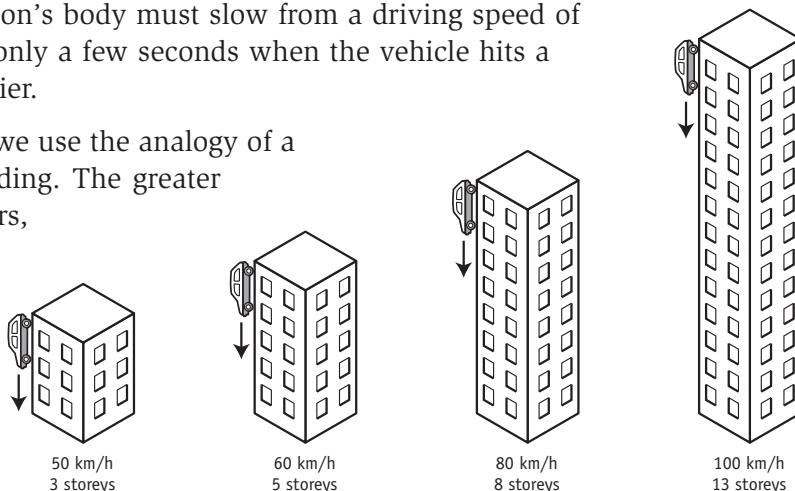




Forces during a collision

During a collision, a person sitting in the vehicle comes under a tremendous force that is very hard to imagine. The person's body must slow from a driving speed of possibly 60 km/h or more to zero in only a few seconds when the vehicle hits a stationary body such as a tree or barrier.

This force is better understood when we use the analogy of a car falling from the top of a tall building. The greater the speed at which the collision occurs, the greater the force needed to stop the car, and the greater the force on the person's body. Hence the higher the building we imagine the car falling from.



The effect of speed on stopping distance

It takes only a few experiments in a car to see that the greater the speed of a vehicle, the longer the stopping distance, and hence the greater the chance of an accident occurring.

When a moving vehicle needs to stop suddenly, the time it takes for it to actually stop depends on two main things:

1. The **reaction time**, which is the time it takes a driver to observe a situation and react to it.
2. The **braking time**, which is the time it takes for the vehicle to actually stop once the brake has been applied. So,

$$\text{Stopping Time} = \text{Reaction Time} + \text{Braking Time}$$

The table below shows the reaction and braking distances for an alert driver driving a new car with excellent brakes on a good road at a range of different speeds.

Speed (km/h)	10	20	30	40	50	60	70	80	90	100	110
Reaction distance (m)	2	4	6	8	10	12	14	16	18	20	22
Braking distance (m)	1	3	6	11	17	24	33	43	54	66	78
Stopping distance (m)											

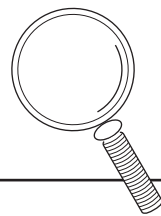


Exercise

1. Complete the last row of the table above by adding the reaction and braking times together.
2. Draw a bar graph of the distances on the grid drawn on the next page. Use one colour (eg red) for the reaction distance, then in a second colour (eg blue) add the braking distance so that the total length of the two coloured bars is equal to the total stopping distance.

Why should there be a speed limit?

2 SPEED AND ALCOHOL

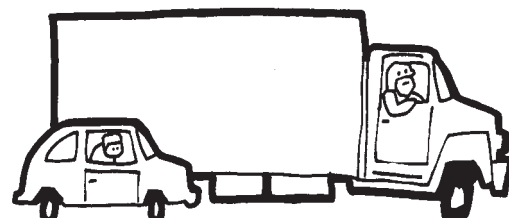
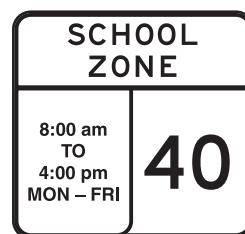
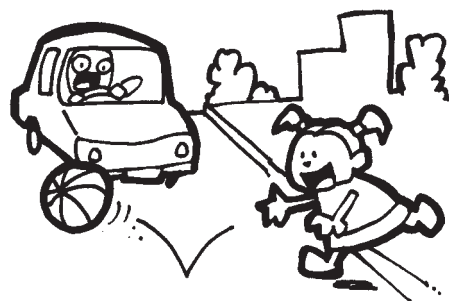


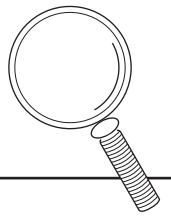
Car Speed (km/h) \ Distance travelled	10 m	20 m	30 m	40 m	50 m	60 m	70 m	80 m	90 m	100m
10										
20										
30										
40										
50										
60										
70										
80										
90										
100										
110										

?

Questions

- What happens to the stopping distance when the speed is doubled from 40 km/h to 80 km/h? Does the stopping distance double or does it increase even more? Explain your answer.
- Explain what happens to the force on a passenger during a collision if the speed of the car is increased from 50 km/h to 100 km/h.
- Imagine you are in an average family car travelling at 60 km/h and a child steps onto a pedestrian crossing when you are 40 metres from the crossing.
 - Would you be able to stop? Describe how close you would be.
 - Describe the difference if it was raining?
 - How different would the stopping distance be if you were travelling at 50 km/h?
- Explain why many local governments have now reduced the legal speed limit in suburbs from 60 km/h to 50 km/h?
- Why do you think that the speed limit on roads located around schools on school days is 25 - 40 km/h, depending on which state or territory it is in? Do you agree with this rule? Explain why or why not.
- The figures in the tables are for an average sized car.
 - How do you think the stopping distance would differ if it were a large, heavily laden truck driving at these speeds? Explain your answer.
 - Do you think the same stopping distances would apply if it were a small 2-door car? Explain.





It is not speed that kills or injures people, but the force of an impact on their bodies when they collide with a stationary object. The greater the speed, the greater the force of the impact, and so the more dangerous it is for the person.

The compulsory wearing of seat belts was first introduced around Australia in the 1970s and further extended in the 1980s. Before seat belt wearing was made compulsory, extensive studies were done comparing the injuries sustained in accidents by drivers who were wearing seat belts and those who were not. The table below shows the results of one such study.

A survey of 20 000 drivers who were involved in car accidents was carried out to see how the wearing of seat belts affected the type of injuries that were sustained.

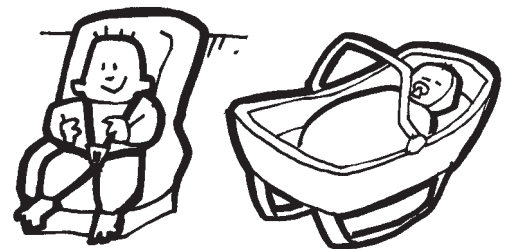
Type of injury	Number of drivers not wearing seat belt	Number of drivers wearing seat belt
Fatal	17	3
Serious	121	75
Slight	387	254
No injury	9475	9668
TOTAL	10 000	10 000

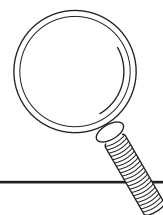
Seatbelt webbing is made from special polyesters that must conform to very strict Australian standards that include strength and ability to stretch under load then recover from the stretch. It must also be resistant to sunlight degradation and abrasion. Tests are carried out to ensure all seatbelt webbing meets these standards.

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Questions

- Imagine you are in charge of making a decision regarding the wearing of seatbelts. Using the figures above, what decision would you make? Give reasons for your decision.
- People who are caught not wearing seat belts are fined over \$100 and the driver may lose up to 3 demerit points. Discuss how fair you think this is. If you disagree with this punishment, suggest what you would do if you were in charge and give reasons for your answer.
- Are there any people in your state or territory who are exempt from wearing a seat belt? Find out what the laws are regarding the wearing of seat belts for the following groups of people:
 - pregnant mothers
 - small toddlers
 - very young babies
- When do seat belts suffer abrasion? What damage can sunlight do to seatbelt webbing? Explain why seatbelt webbing must be resistant to both abrasion and damage from sunlight.
- Describe how an Australian Standards laboratory might test the strength of the fibres in seatbelt webbing. What mass do you think it should be able to hold?
- What would you say to someone who argues that it is our democratic right to decide for ourselves whether to wear a seat belt or risk injury?

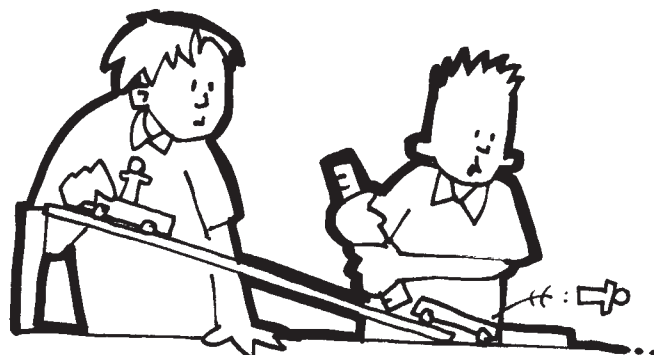




In this activity you are going to investigate what happens when a car and passengers are involved in a collision.

Materials

- 2 small trolleys
- 2 plasticine passengers
- some masking or sticky tape
- a wooden ramp about 1-2 m long
- a 40 cm or 1 metre ruler



Procedure

1. Rule up a table of *Results* in your science book, as shown below.

Record distance, in cm, and direction (forwards {F}, backwards {B}) passengers A & B are thrown.

Test	Height of ramp (cm)	Run 1	Run 2	Run 3	Average (cm)
		A , B	A , B	A , B	
slow	30	,	,	,	,
fast	40	,	,	,	,
slow / tape	30	,	,	,	,

2. Set up trolley A with its 'passenger' at the top of the ramp, and trolley B and its 'passenger' stationary at the bottom.
3. Release trolley A and observe what happens to both passengers when the trolleys collide. Measure and record the distances and directions in which they fell. Repeat twice more.
4. Now raise the ramp using a further book or brick. This will increase the speed of the trolley. Repeat the same steps three times and record your results.
5. Lastly, lower the ramp height back to the original height, but this time use masking tape to fix both passengers to their trolley. Repeat the collision three times and record your results.

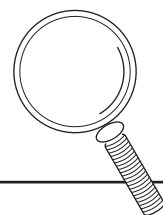
Questions

1. Explain why each test was carried out three times and then the average calculated?
2. Describe what happened to the average distance the passengers were thrown out when the speed was increased? Did the direction that they fell change? Explain your answer.
3. What do you think the masking tape represented?
4. What happened to the passengers when the masking tape was used? Explain why this occurred.
5. How can the results from this experiment be used to argue that it is safer to wear seat belts?
6. What did you notice about the direction each passenger moved after the collision? Explain why this happens.

Conclusion

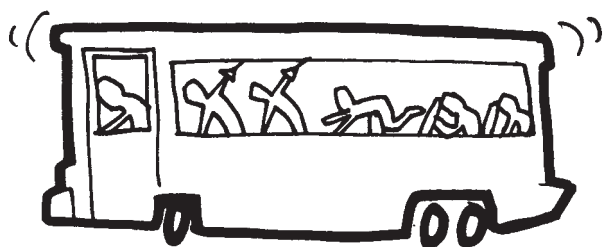
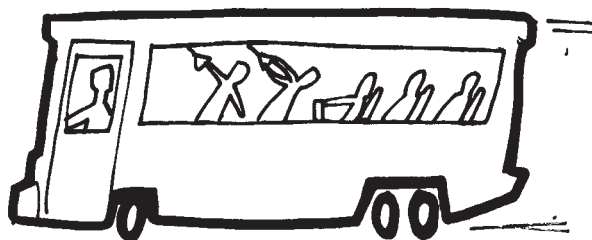
What can you conclude about the seriousness of injuries to passengers when

- (a) the speed of the vehicles is increased?
- (b) seat belts are worn?



The reason why a person keeps moving forward in a collision is the property of **inertia**. It means that a body will continue moving in the direction it was going, or will remain at rest, unless it is acted on by a force. This is also called **Newton's First Law of Motion**.

This property is clearly seen if you are standing up in a stationary bus. If the bus takes off suddenly, your feet, which are attached to the bus floor, will move forward with the bus because this force is acting on your feet. However, the rest of your body will try to remain behind where you were and you feel as if you are falling backwards.

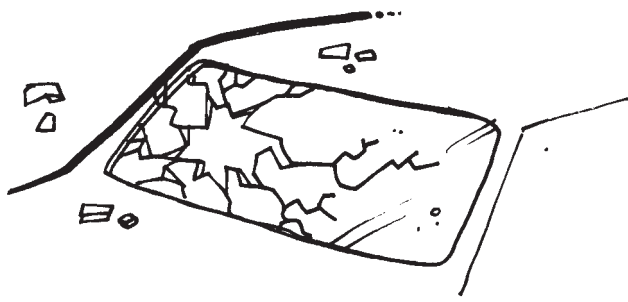
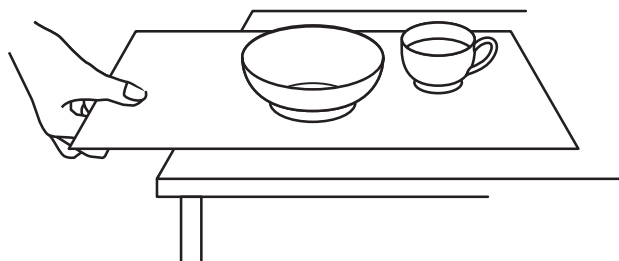


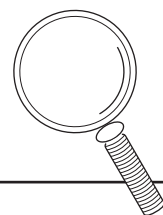
Now if the bus stops suddenly, again your feet, which are in contact with the bus floor will stop, but the rest of your body will attempt to continue moving forward at the same speed, until your legs, which are attached to your feet, pull your body back. This means that you feel as if you are falling forwards.

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Questions

1. Why do you think all new cars now have headrests above the actual seats? Explain how they should be adjusted and why.
2. Explain how whiplash injuries might occur. What can be done to avoid this injury?
3. Discuss why station wagon owners who carry heavy loads in the backs of their cars are advised to have a strong grill between the back section and passenger compartment.
4. Some magicians are able to do a 'trick' in which they give a tablecloth a very quick tug such that the tablecloth is pulled out from under the dishes, which are left standing in their original places on the table. Use the concept of *inertia* to explain how this is done.
5. Describe, using the word 'inertia', what happens when a passenger not wearing a seat belt is in a car that collides with a tree. Explain the difference when the passenger is wearing a seat belt.
6. Look at the picture of a front windscreen on the right. What information can the Accident Investigation Team present at an inquiry into the death of the driver?





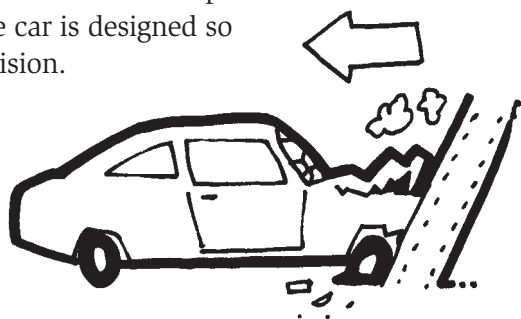
While compulsory seat belts have had a big effect on reducing the road toll, it was felt that even more could be done. Over recent years two new technologies have been developed to give additional protection to people who find themselves involved in a road accident. What was needed was for the body to be cushioned so that it could come to rest a little more slowly.

An air bag inflates in only a fraction of a second so that a person moving forward into their seat belt is cushioned before their head can reach the hard obstructions in front of them.

The other new feature that most new cars now have is what is called a 'crumple zone'. This means that during a collision, the front part of the car is designed so it will slowly crumple up, taking much of the force of the collision.

The passengers are held safely in a stronger compartment that stays intact but will come to rest more slowly as the front of the car takes much of the force.

The following activity will help you understand how the idea of a crumple zone works.

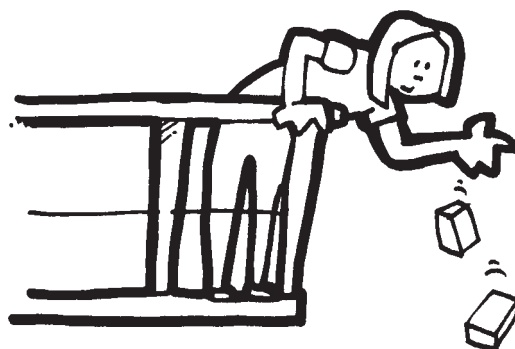
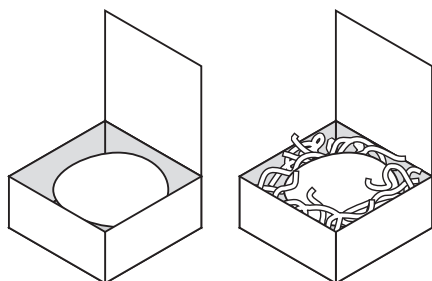


Materials

- 2 fresh eggs
- 2 cardboard rolls or small boxes the eggs will fit into
- some of waste paper
- sticky tape

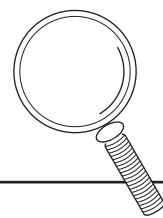
Procedure

1. Place the first egg into one of the cardboard rolls or boxes and place the second egg into an identical container, but this time make sure there is a lot of crumpled paper in front of the egg to stop it hitting the sides of the cardboard.
2. Drop both containers from a balcony that is at least 3 metres off the ground. When both have fallen open the containers to observe the results.



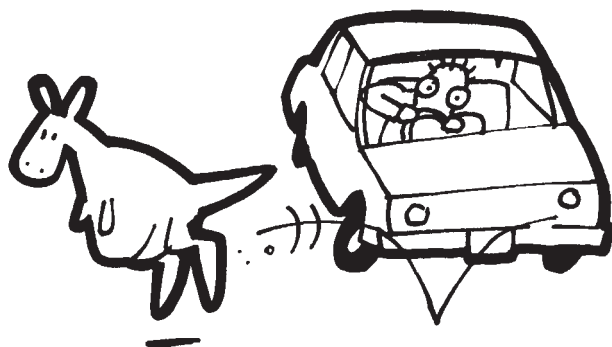
Questions

1. What did you notice about the conditions of the two eggs after they had dropped from the balcony? Explain why this difference occurred.
2. What do you think the following things represented?
 - (a) the eggs
 - (b) the crumpled paper
 - (c) the cardboard box or roll
3. Try to think of two other situations that you know of where the idea of a 'crumple zone' is used to cushion something or someone to reduce the force of an impact.
4. Explain what is meant by the word 'control', when running an experiment. What was the control you used in this experiment?



Factors affecting reaction time

In unexpected situations, such as the car in front braking suddenly or a child or animal running out onto the road, it takes some time for the driver to react to the danger. The situation must first be observed, then it takes a certain amount of time for the message to reach the brain, where it is processed, and then another message is sent to the muscles of the leg to contract and push on the brake.



The time from when the situation first occurred, up to the time the brakes were applied, is called the reaction time. This time can be affected by a number of things, including tiredness and hence the alertness of the driver, being under the influence of alcohol or other drugs, or being distracted by things either inside the vehicle or outside it.

Calculating reaction distance

By rearranging the formula for calculating speed ($v = \frac{d}{t}$), the distance travelled can be calculated if the speed is known.

ie **reaction distance** = time taken to react $\times$ average speed

It has been measured under test conditions, that the average reaction time for an alert driver is about 0.75 seconds. However, in normal situations on the road, a driver will not be as alert as when he or she knows he or she is being tested. Hence it is more realistic to assume a reaction time of about two or three seconds.

Now using the formula above for calculating distance, we can easily work out how far the car will travel in the time it takes the driver to react. For example:

What is the distance a car will move, while travelling at 60 km/h, when an alert driver takes one second to react to a child who has run out onto the road?

Note: It is best to convert km/h to m/s (divide it by 3.6) so that calculations are easier and the final distance will be in metres.

$$\begin{aligned}\text{distance} &= \text{time} \times \text{speed} \\ &= 1 \text{ second} \times 60 \div 3.6 \text{ m/s} \\ &= 16.7 \text{ metres}\end{aligned}$$

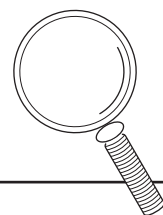
Summary

Reaction distance depends on the alertness of the driver and the speed at which the car is travelling.

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Questions

1. Calculate the reaction distance of a distracted driver who is travelling at 100 km/h, and takes 3 seconds to react to an accident up ahead?
2. How far will a vehicle travel if its driver takes $1\frac{1}{2}$ seconds to react to a traffic light that has changed to red, when the vehicle was moving at 60 km/h?
3. How much longer is the reaction distance if the car in question 2 above was travelling at 80 km/h?
4. Describe how the reaction distance could be affected if the driver was under the influence of drugs.



In this activity you will investigate what sort of things might affect the time it takes a person to react to an unexpected situation.



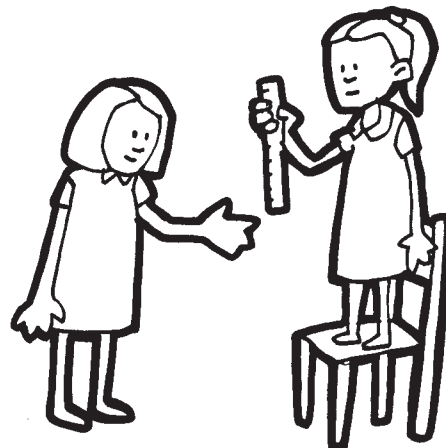
Materials

- a 30 or 40 cm ruler
- three student volunteers per group



Procedure

The first student (volunteer 1) stands on the floor with their hand near the zero mark and 5 cm away from the ruler, which is held by the second student (volunteer 2) who is standing on a chair as shown on the diagram. When volunteer 2 lets the ruler go, volunteer 1 must try to grip it as quickly as possible. The longer the reaction time, the higher the reading on the ruler.



The table below should be ruled up in your science book under the heading 'Comparing Reaction Times'.

Test	Run 1	Run 2	Run 3	Average (cm)
1. Hand held 5 cm from ruler				
2. Hand held 10 cm from ruler				
3. Volunteer being distracted				

The test should be repeated twice more so that an average result can be calculated. Does the volunteer improve with practice?

The same test should now be done three more times but with the hand 10 cm from the ruler, and a further three times with the hand 5 cm from the ruler but with a third volunteer trying to distract the first student. If you have time repeat using different students.



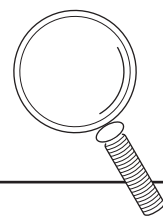
Questions

1. Did the first student's reaction time change over time? That is, did practice improve the reaction time?
2. How was the reaction time affected when the person's hand was further from the ruler?
3. Describe how the reaction time was affected when the student catching the ruler was being distracted.
4. When comparing the average reaction time for several student volunteers, did some people have consistently shorter reaction times than others? If so, why do you think this is?
5. How do you think a driver's reaction time would be affected if the person was very tired or under the influence of mind-altering drugs?



Conclusion

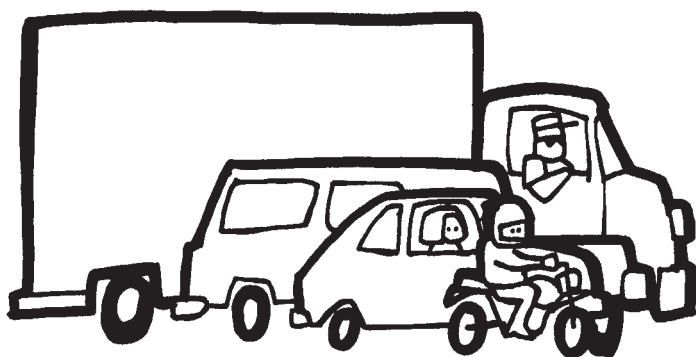
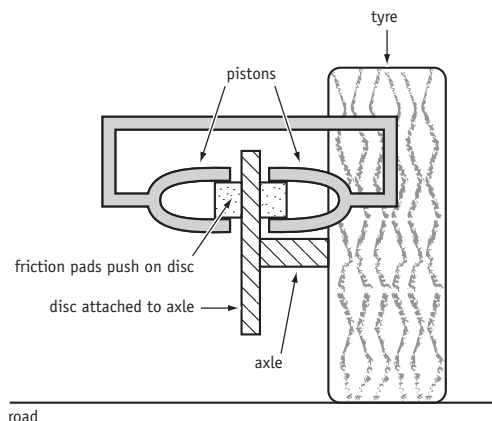
What has this activity taught you about factors affecting reaction time?



Friction

When brakes are applied, they rely on the grip of the tyres on the road and the grip of the brake pads on the tyres to bring the vehicle to a stop. Another word for grip is **friction**, which is a force that resists movement. Friction depends on the roughness of the two surfaces that are in contact. The rougher the surfaces, the greater the friction or grip.

Besides friction, two other factors that affect a car's ability to brake are its **mass** and **speed**. Generally, the heavier a vehicle is, the more force that is needed to bring it to a stop, and usually the longer it will take for this to occur.



This can easily be understood if you imagine a large truck, a medium car, a small car and a motorcycle all travelling at exactly the same speed. If they all have to stop suddenly, it will take a much greater force to stop the large truck than the smaller vehicles. Now if the same force is used, then it will take a much longer distance to stop the truck than a car or motorcycle.

Similarly, if two identical vehicles are travelling at different speeds, one at 20 km/h, the other at 60 km/h, then it is easy to understand that the faster car will take longer, or need a greater braking force, to bring it to a stop.

This property, combining speed and mass, is called **momentum**.

$$\text{Momentum} = \text{mass} \times \text{speed}$$

In most cases the exact mass of a car is not known. It has been found by testing, that an average family sized car with good brakes, travelling on a dry road, will be able to reduce its speed by 7 m/s to 8 m/s every second. This means that a car travelling at 60 km/h will take just over two seconds to stop, after the brakes are applied.

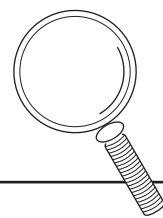
Calculating braking distance

To calculate the braking distance, remember that:

$$\text{distance} = \text{average speed} \times \text{time}$$

To calculate the **average** speed, find the **mean** of the initial speed and the final speed (zero).

$$\text{average speed} = \frac{\text{initial speed} + \text{final speed}}{2}$$

**Example**

Find the braking distance of an average family car with good brakes, travelling at 60 km/h on a dry bitumen road. From the information on the previous page, assume that the braking time is 2.5 seconds.

The average speed = $(60 + 0) \div 2 = 30$ km/h

So the braking distance = $30 \text{ km/h} \times 2.5 \text{ seconds}$ (*change 30 km/h to m/s by $\div 3.6$*)
 $= 8.3 \text{ m/s} \times 2.5 \text{ seconds}$
 $= 20.8 \text{ m}$

Summary

Braking distance depends on the speed and mass of the car and the conditions of the tyres, brakes and road surface.

X**Exercise**

- In your science books, write the heading '*Increasing Friction*'.
 - Make a list of as many ways you can think of in which friction can be **increased**.
 - Give four examples of situations where friction is needed and must be as great as possible.
- Write the heading '*Decreasing Friction*'.

- List as many ways you can think of to **decrease** the effect of friction.
- Discuss four situations in which friction is a nuisance and needs to be minimised, discussing how the friction is reduced.

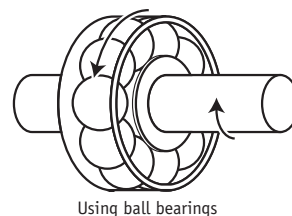
- You have probably found that all the different ways in which friction can be reduced fall into three main groups:

- reducing the amount of surface that is in contact, as in rollers or ball-bearings;
- using a lubricant such as oil or grease; and
- making the surfaces as smooth as possible so that they can slide past each other.

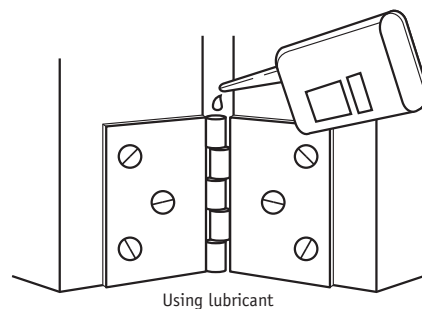
Now transferring the theory above to the road situation, describe the effect you think the following things would have on friction and hence stopping distance.

- | | |
|------------------------|------------------------------|
| (a) a gravel road | (d) new, extra grip tyres |
| (b) bald or worn tyres | (e) an oil spill on the road |
| (c) rain | (f) worn out brake linings |

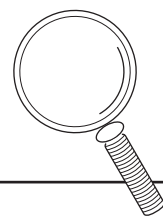
- Design an experiment that tests the effect of all six of the above situations on stopping distance. List the equipment you would need and the method you would use to test the hypothesis made in the previous question.
- What would be the braking time of a new car travelling at 100 km/h on a dry road? How far would the car move in this time?
- Calculate the braking distance of an old car doing 90 km/h with worn brakes, such that it can only reduce its speed by 5 m/s every second. (*Hint: work out braking time first*)
- What is the difference in braking distance if a new car in good condition does 120 km/h instead of 100 km/h on a dry bitumen road?



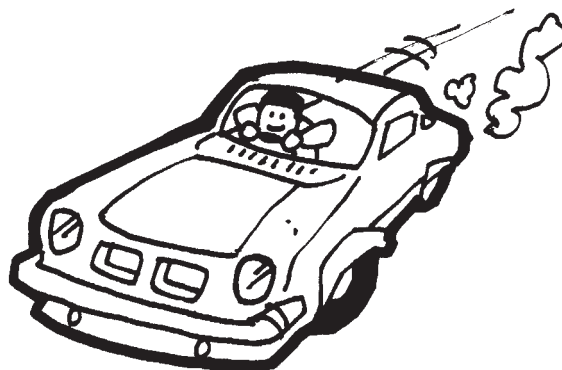
Using ball bearings



Using lubricant



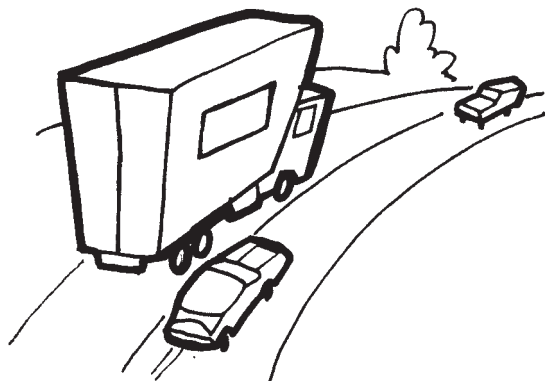
It was a Friday evening when the fatal accident occurred. Andy Allen was an Agricultural Chemicals salesman who spent a lot of his work time travelling throughout the countryside visiting farmers with samples of his firm's product. His job was to persuade farmers to buy his company's drenches, feed supplements and other products. His car was a recent model and he appreciated its comfort as he spent a lot of his day sitting in it as he travelled from farm to farm.



Andy had been away from home the whole week. He was tired and looking forward to getting home to spend the weekend with his wife and young daughter. It had not been a good week. The farmers were not keen to spend money trying new products and, on top of that, he never slept well in strange motels. As he was driving towards home, he was thinking about what he could do next week to try to improve his sales figures because his boss had been hassling him about it.

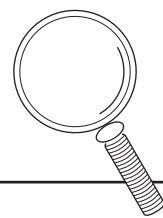
The road he was travelling on was a major one between two large towns. The road surface was quite new and Andy was making good time. He had only about another 100 kilometres to go, so estimated that he would be home in about another hour. He decided to ring his wife on his mobile phone to let her know the time he expected to be home. With annoyance and impatience he found that he was catching up to a large slow truck on the hills. As well as that, it had just started to drizzle, so he had to put his windscreen wipers on.

Belinda Brown was on her way home on the same road, but travelling in the opposite direction. She was thankful that it was Friday. She worked in the local solicitor's office in the large town that she had just left behind. She lived with her husband and four-year-old son on a small property about 20 kilometres out of town. It was 6.30 pm, and just starting to get dark. Jason, having been picked up from childcare, was whining from the back seat, as he was tired and hungry. Belinda now regretted staying after work to have a few celebratory drinks with the rest of the office staff, as it had made her late and she had nothing organised for dinner. She was also a bit concerned that she had been talked into having a second glass of champagne. She felt alert, but knew that if the police tested her she could be over the 0.05% alcohol limit. She would just have to be extra careful not to do anything that would attract their attention. As the rain started, she automatically turned on her wipers and for the third time told Jason to sit down in his seat so she could see out through her rear vision mirror. She was feeling stressed and angry.



Just at the time that she was getting cross with Jason, Andy caught up to the truck, which was slowing down as it was driving up another hill. The road ahead looked clear, so Andy accelerated to 120 km/h and moved out to overtake the slow-moving truck.

As Belinda looked forward again, she reached the top of the hill and saw the large truck coming towards her, with Andy's car in her lane driving directly at her. She pressed hard on the brakes and tried to avoid the car and truck, but on the wet road the car's wheels locked and she slid sideways into the gravel at the side, and into the oncoming car, which had also veered into the gravel to avoid her.



Belinda was killed instantly. Her son Jason, who was not wearing a seat belt, was thrown out of the unlocked car door and severely injured. The truck driver was treated for shock at the scene but was uninjured. He had radioed for an ambulance to come from the nearest town.

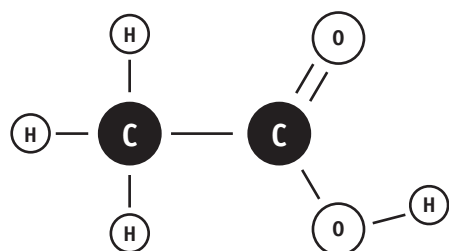
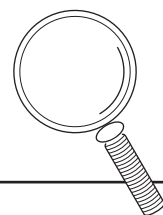


Andy's late model car was fitted with anti-locking brakes which helped him keep the car straight, but he was unable to avoid the collision when he and Belinda had veered to the same side of the road. He was badly hurt, but the airbag that inflated in a millisecond had probably saved his life. He had chest injuries and was trapped between the seat and engine, which had been pushed forward onto his legs. There was some talk about amputation, but after about half an hour he was freed by the fire brigade and ambulance officers.

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Questions

1. Do you think this accident could have been avoided, or was it just bad luck? Who do you consider was most at fault? Explain why.
2. Discuss all the things that Andy could have done to have avoided this collision.
3. Did Belinda contribute in any way to this accident? If so, in what way?
4. What things did Belinda fail to do as a responsible parent? Can you explain why this sometimes happens? Suggest ways in which these problems can be overcome to ensure everyone in the car is as safe as possible when travelling.
5. Find out how anti-locking brakes work. If Belinda's car had been fitted with these, is it possible that she might have been able to avoid the collision?
6. How might the weather have contributed to the accident? What should responsible drivers do when it starts to rain?
7. The police told Belinda's husband that the two back tyres of Belinda's car should have been replaced. Why would they have commented on this? How could the condition of the car's tyres have contributed to the accident? Find out the amount of tread that a tyre must have to be legal.
8. If Belinda's car had been fitted with air bags, how do you think this would have changed the way she and her son came out of the accident?
9. The Accident Investigation Team arrived at the scene only minutes after the ambulance and fire brigade, and these people were already helping the victims. Make a list of all the things these officers would do to collect enough evidence to determine the cause of the accident.
10. Do you think that Andy should be convicted of any charge? If so, which one(s)? Explain the reasons for your answer.



Ethanol molecule
C = carbon, O = oxygen, H = hydrogen

The substance we commonly call alcohol is the chemical **ethyl alcohol** or **ethanol**. Its chemical formula is CH_3COOH , and it is found in nature in fermented fruit. Fermentation is where sugars in the fruit, combined with natural fungi like yeast, cause a chemical reaction, which produces ethyl alcohol and carbon dioxide gas.

Alcohol is a depressant. It slows a person's reaction time and can distort his or her judgement of speed and distance. It also reduces a driver's ability to concentrate.

The more a person drinks, the greater the effect. For this reason, throughout Australia, if a person has a full driver's licence, the maximum blood alcohol concentration (BAC) is 0.05%. This means that there is 0.05 gram of alcohol in every 100 millilitres of blood.

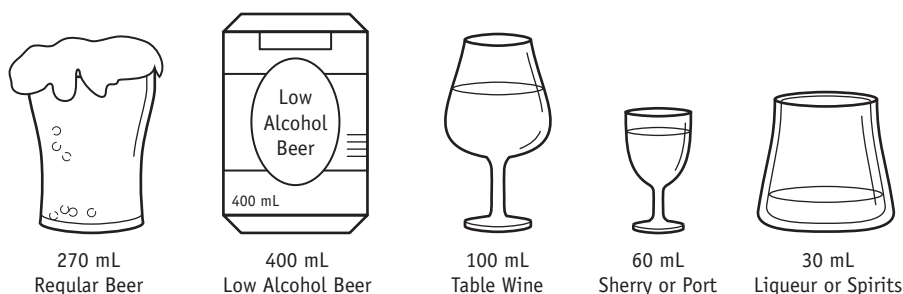
Once alcohol is in the body, it can only be removed by being broken down in the liver. This process takes time. The liver can only break down about 8 grams of pure alcohol, or one standard drink, per hour. Nothing, including drinking strong black coffee, cold showers or sleeping it off, can speed up this process. This means that once a person has reached the legal limit, consuming one standard drink per hour will keep them at that level.

A person's level of alcohol is affected by both their weight and their sex. The lighter a person is, the less alcohol is needed to put them over 0.05. Also, if the person has not eaten recently, because alcohol is absorbed more quickly if the stomach is empty, the effect may be greater.

As a general guide, women should think carefully before their second drink, because two standard drinks in the first hour may put many women over the legal limit. Men are likely to go over 0.05 if they have a third drink in their first hour.

What is a standard drink?

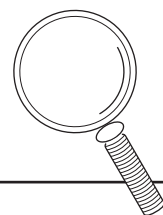
Any of the following are considered a **standard drink**:



Each of these different volumes of alcoholic drinks contains approximately 8 grams of alcohol.

The effect of alcohol on the body

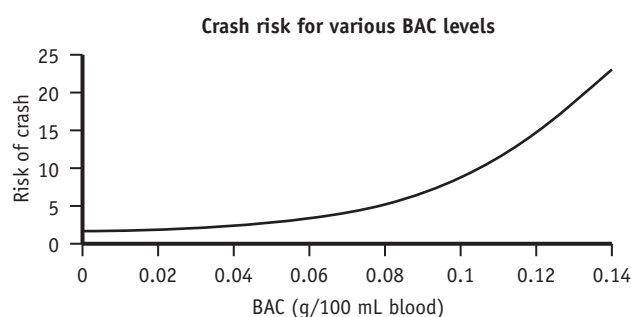
In the body alcohol is absorbed from the stomach and small intestine and into the blood stream. In fact it can be detected in the blood within five minutes of being consumed. The blood carries the alcohol to the liver where it is first changed into the poisonous ethanal (acetaldehyde), then into ethanoic acid (acetic acid). This substance is one of the fatty acids that can be carried around the body and absorbed into different body cells. Here it can be broken down into carbon dioxide and water. The process is called cell respiration and releases energy. Naturally, if the energy produced exceeds what the body needs, energy reserves, in the form of fat, are laid down as a form of storage.



The immediate effect of alcohol in the blood is that it slows the brain's function. The first parts of the brain affected are those that control the emotions and behaviour. Some people may exhibit behaviour that would be uncharacteristic for that person. Others become more aggressive as they are unable to control their anger; or a shy person may become less inhibited and may be more socially interactive. Another person may become quite morose or depressed. Judgement of speed and distance is often affected, which makes it dangerous for such people to be in control of a car.

The risk of being killed or seriously injured in a car crash when the driver has been drinking has been calculated from accident statistics. It is found that when the BAC of the driver is doubled, the risk is not doubled, but increases at a much greater rate. This is seen in the graph below. Note that at a BAC of 0.05 the risk of having a serious accident is $2\frac{1}{2}$ times greater than if no alcohol has been consumed. If the BAC is doubled to 0.1, the risk of causing an accident is increased 8 times.

If a greater amount of alcohol is consumed, more of the brain will be affected. Speech may become slurred and vision blurred. The drinker may experience loss of memory. Alcohol can also produce tunnel vision, where the person is no longer aware of the 180° field of vision. Balance may be lost, and any further alcohol will cause most people to lose consciousness. In extreme cases, death may follow.

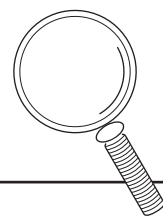


People who drink excessive amounts of alcohol regularly can cause permanent damage to their brain, liver, stomach lining, nerve tissue and heart.

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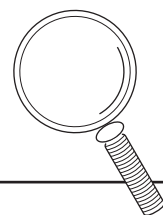
Questions

1. What is meant by BAC? What is the level of blood alcohol that a driver on a full licence is legally allowed? At what level do you think it is safe to drive? Explain why.
2. Explain what is mean by the term 'standard drink'? Approximately how many grams of alcohol is in a standard drink?
3. About how many standard drinks in the first hour, will put an average male driver over the legal limit? How does this differ for females?
4.
 - (a) Explain how a person can get their BAC back below 0.05.
 - (b) Which organ in the body is responsible for breaking down the alcohol?
 - (c) How much alcohol can this organ break down in an hour?
 - (d) Which part of the body is most quickly affected by alcohol?
5. Jane and Bill went out. In three hours Jane drank two glasses of table wine and a mixed drink containing 30 mL of a liqueur. Bill drank four glasses of 200 mL of low alcohol beer and a port.
 - (a) Approximately how many standard drinks did each person have?
 - (b) Which person is probably the fittest to drive home?
 - (c) Is either person likely to be over the legal limit, if both had full licences?

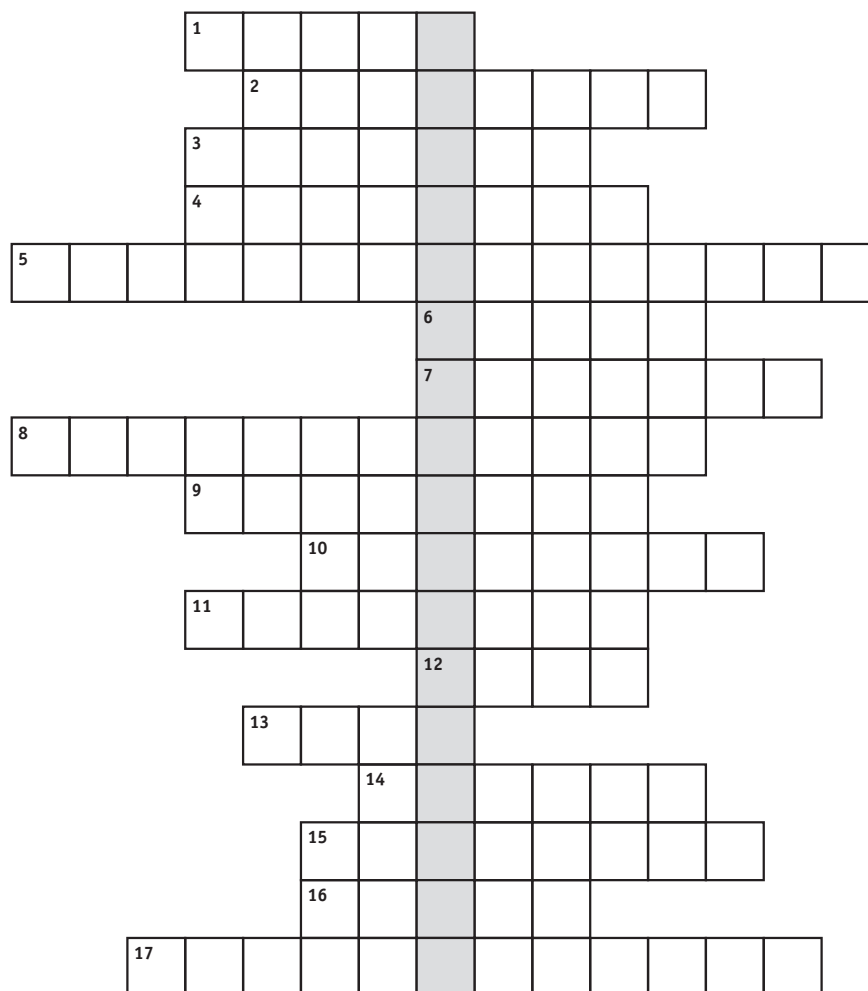


1. (a) What is meant by the words **speed** or **velocity**?
(b) In what two units is speed usually measured?
2. (a) Calculate the average speed of a car that travels 350 kilometres in 5 hours.
(b) What is the average speed of a sprinter who runs 200 metres in 25 seconds?
(c) Convert the speed in question 2b into km/h.
3. (a) What is meant by the term **inertia**?
(b) List three safety devices fitted to cars to prevent injuries caused by the effects of inertia in a collision.
4. (a) What is meant by the term **momentum**?
(b) What are the two factors that affect momentum?
5. (a) What two components combine to give the total **stopping distance** of a vehicle?
(b) Make a list of five things that can **increase** a driver's reaction time.
(c) What are five things that can **increase** a car's braking time?
6. (a) Calculate the reaction distance if a driver travelling at 75 km/h takes $1\frac{1}{2}$ seconds to react.
(b) Calculate the braking time of a car that is travelling at 90 km/h and is able to reduce its speed by 15 km/h every second.
7. (a) Explain why alcohol is classified as a drug.
(b) Describe at least six side-effects of drinking alcohol that can have a detrimental effect on a person's ability to drive.
8. (a) Until recently, some Australian states had a BAC limit of 0.08. Explain why those states have now joined the rest of Australia to make the national legal limit 0.05.
(b) How does this level differ for: (i) a Learner's licence?
(ii) a Provisional licence?
(c) Why would drivers holding a Learner's or Provisional licence have a lower limit than a driver on a full licence?
9. What advice would you give to a person whom you suspect is over the legal limit and is preparing to drive home?
10. For each of the following drinks give its quantity in numbers of standard drinks.
(a) 400 mL can low alcohol beer (d) middy (285 mL) of full strength beer
(b) schooner (425 mL) of regular beer (e) nip (30 mL) of spirits
(c) small glass (120 mL) of port (f) 120 mL glass of red wine
11. How many standard drinks can a female, on a full licence, drink in the first hour and still remain under a BAC of 0.05?
12. Explain why cold showers and black coffee do not reduce the amount of alcohol in a person's blood. What is the only method by which blood alcohol level can be reduced?



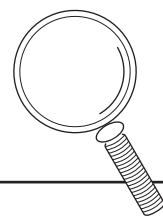


Find the statement hidden in the shaded column.



Clues

1. This is the cause of far too many accidents.
2. Another word for speed.
3. Scientific name for alcohol.
4. We need a lot of this for efficient braking.
5. The distance a car travels while the brakes are applied.
6. If a surface is this, there is good grip.
7. The property of an object that resists a change in motion.
8. The time from when a driver sees a danger, to when the brakes are applied.
9. This does not mix with driving.
10. This safety device prevents injury in an accident due to inertia.
11. The measure of mass multiplied by speed.
12. When the road is this, a driver should take care and reduce speed.
13. If the wheels lock, the car will do this.
14. These should be kept in good order to shorten stopping distance.
15. A type of injury caused to the neck, if the head is not restrained in a rear collision.
16. The organ of the body that breaks down alcohol.
17. An increase in speed.



1 Blood and DNA

1-1 What is blood?

1. 5-6 litres of blood
2. A person's blood group; DNA profile and hence sex; possibly degree of injuries and how blood was spilt.
3. Hopefully the sequence of events and details about how the attack was carried out.
4. Plasma – carries dissolved substances; Red blood cells – carry oxygen around body to cells; White blood cells – fight disease; Platelets – help blood to clot.

1-2 Blood groups

1. The ABO blood group system.
2. O blood group
3. This suspect may have committed the crime.
4. AB group because it is the rarest group.
5. Around Mongolia and north Asia, spread by migration.
6. A secretor is someone whose blood cells leak into other body fluids, so blood details can be found from other biological fluids.
7. $A+ = 33\%$ (ie $0.39 \times 0.82 = 0.3298$); $AB- = 0.007\%$; $B- = 0.02\%$
8. Not enough evidence to prove guilt, but it is strong evidence.
 $O+$ blood would be less useful because it is very common.
9. Such a person is likely to die because the foreign antibody will cause the red blood cells to clump together so that they can no longer carry out their task.
10. If a pregnant mother is Rh- and the father is Rh+ then problems may occur if an unborn baby is Rh+. To avoid the death of future children, the mother will be injected with a substance that will prevent her body producing antibodies that may attack the blood of any future Rh+ babies.

1-3 Demonstration: Identifying blood

1. Pink in KM, green in LMG and hemastix
2. Tests are very accurate even if only a little blood is tested.
3. Tests are very useful; quick and easy to use. Few substances give false positives.
4. CSE would wear gloves and a mask to do a presumptive blood test then collect the sample on a moist swab and place it in a sealed container.
5. Label carefully and send to forensic lab.
6. It would be tested to see if the blood was human or from another animal using the Uhlenhuth Test. If human, it would be DNA tested, and maybe blood-typed.

1-4 Blood splashes

1. If the body was moved before or after death; height blood dropped from; if the victim was moving; direction and speed of impact; type of weapon.

2. (a) Dropped from a height of 1-2 m.
(b) The victim moved or the blood fell on sloping surface.
(c) The blood was smeared.
3. Cast offs are very small blood droplets released from a weapon during an attack in which the victim is hit by several blows. They may tell which hand the assailant used, how many swings were made, how the weapon was used, and the position of the assailant and victim during the attack.
4. Artery damage will leave a lot of blood and patterns not unlike that left by water from a garden hose.
5. (a) Take care that nobody (self, witnesses etc) is in any further danger.
(b) Check health of victim and help with first aid if appropriate.
(c) Call ambulance.
(d) Secure crime scene to prevent further contamination of evidence.
(e) Make a quick initial assessment regarding what might have occurred.
(f) Get names and addresses of any witnesses.

1-5 Activity: Blood splash testing

1. Size of spots, length of spines.
2. The steeper the slope the narrower and more elongated the oval shape of the splash.
3. Movement causes the shape to be less round and more elongated, often with small secondary spots in the direction of the movement. The faster the movement the narrower the shape and the further the distance between spots. Knowing the speed and direction of a bleeding person could help reconstruct the crime.
4. The thicker the blood the rounder the spots in the case of movement, and the shorter the spines. This could be tested with thick paint and some paint diluted with water.
5. Blood becomes thick when it starts to clot.
6. The more information available, the greater the confidence with which the crime can be reconstructed.

1-6 Case study

1. Bloodstains on the victim's pillow and a bloody imprint were thought to have come from a surgical instrument and so the doctor was thought to be guilty.
2. No blood on ceiling; cast offs on walls; doctor was right-handed, while attacker used left hand; only a small amount of blood on doctor. This led to the conclusion that the doctor probably did not carry out the murder and he was freed.
3. Fingerprinting ransacked room; examination of sofa downstairs for doctor's head hair.
4. Additional evidence increases the accuracy of findings.
5. The crime scene was not searched thoroughly enough immediately after the attack.

1-7 What is DeoxyriboNucleic Acid?

1. In the nucleus of every body cell that has a nucleus. It carries the genetic code that is passed on from parent to offspring.
2. The DNA molecule splits in half like a zipper so that each half can be copied to make it complete again, forming two new identical DNA molecules.
3. DNA is unique for each combination of sperm and egg, so the only people to have the same DNA are identical twins who grew from the same egg and sperm. Both fingerprints and DNA profiles are now stored on national databases. Many criminals take precautions not to leave fingerprints at a crime scene, but they are not as easily able to avoid leaving biological material from which a DNA profile can be made.
4. 'Genetic Code' refers to the sequence or order of the bases on the DNA molecule that is a 'code' or 'instruction' for different physical characteristics.

1-8 Activity: Extracting DNA from peas

1. Write the Method in past tense listing all the steps that were carried out.
2. The DNA should be a colourless, almost clear, jelly-like substance.
3. DNA is easily lost when decanting the supernatant in steps 3 and 5, and when extracting 2 mL from the middle of the supernatant in step 8.

1-9 DNA profiling

1. Prof. Sir Alec Jeffreys in 1984 in Leicester University, England.
2. The original method needed a large sample of DNA, it needed to be fresh, the process took a long time and was expensive.
3. Now much less DNA is needed, it is faster and so cheaper. Old DNA can be used in some techniques.
4. It has been reduced from approximately 6 weeks to 2 days.
5. The DNA database contains records of the DNA profiles of previously convicted criminals and from unsolved crimes. When a new person is convicted, or DNA is found at a crime scene, the profiles from these are searched against all other profiles on the database. If no match is found, the new DNA is then added to the database.
6. A large number of previously unsolved crimes have now been solved around the world since the introduction of databases because records can be kept indefinitely, and important invisible biological matter is left behind at many serious crimes.
7. DNA profiling is used to settle paternity disputes and to prove family links in immigration and refugee applications.
8. Blood is an excellent source of DNA from which to make a profile and match it to the profile of a suspect.
9. Blood, semen, vaginal secretions, cheek cells, hair roots, saliva, skin cells, urine and faeces all contain body cells with DNA in their nuclei. The first four substances contain more of such cells.

1-10 Blood and DNA revision questions

1. (a) A. red blood cell; B. white blood cell
(b) A is more common.
(c) A is used to identify a person's blood group.
2. Blood carries oxygen to all cells, and carries away waste products. White blood cells in blood fight disease.
3. (a) 'Class characteristics' are characteristics shared by a group of people.
(b) Shoe prints, layers of paint on a paint chip from a new car or fibres from a garment would all have 'class characteristics' because they are shared by many people.
(c) No individual has their own unique blood group. All people belong to one of the 288 main and minor blood groups.
4. A, B, AB, O
5. Discovered by Karl Landsteiner in Austria in 1901.
6. A group has A antigens, B has B antigens, AB has both A and B antigens, and O has no antigens.
7. (a) O+ is most common at 38%.
(b) AB- is least common at 0.07%.
(c) People who are genetically pure Australian Aboriginal do not have B antigens, so do not belong to either B or AB groups.
8. Blood group is inherited and so determined at conception. It never changes.
9. (a) O group can be given to people from other groups because it does not have any antigens.
(b) People with AB type blood can receive any other blood type because they already have both antigens.
10. M and N blood grouping factors have been found as well as many others.
11. A total of 288 different groups are possible.
12. Only O type blood is possible, as both parents have no antigens to pass on.
13. If one parent has A antigens and the other has B antigens then all four blood groups are possible in their children, including O type blood.
14. (a) The suspect may have been at the crime scene but 38% of the general population has this blood type, so it is not strong evidence.
(b) AB- is much rarer (0.07% of population) so this increases the probability that the blood found at the crime scene came from the suspect.
15. Phenolphthalein goes pink and leucomalachite green goes green in the presence of blood.
16. Rust and some plant enzymes, such as those found in horseradish, will give positive results, but these are unlikely to be found at most crime scenes. The presumptive blood test is quick and easy and if negative the stain can be ignored by the CSE.
17. Investigator can estimate the height blood fell from, position of victim, whether victim was moving, in what direction and how fast.
18. Approximate time of death, whether body was moved before or after death, position of body

during assault, degree victim was able to defend himself.

19. A. Movement to the left. B. Cast offs on a sloping wall.
20. (a) Part of DNA molecule
(b) S = sugar, P = phosphate
(c) A = adenine, C = cytosine, G = guanine, T = thymine
21. Blood, semen, vaginal secretions, cheek cells, hair roots, saliva, skin cells.
22. DNA profiling now allows a person's DNA details to be stored on a database and later matched to the smallest amount of biological matter left at a crime scene.
23. DNA is unique to every individual except identical twins who share the same profile because they came from the same egg and sperm.
24. Father 1 is the real father because he shares 50% of the same DNA with the child.
25. DNA from Suspect 1 has been left at the crime scene. Some DNA from the victim appears there as well. There is no DNA evidence from Suspects 2 and 3 found at the crime scene.

1-11 Blood and DNA wordsearch puzzle

Sentence: DNA profiling is a new discovery.

2 Speed and Alcohol

2-1 Traffic accidents

1. To gather evidence so that the cause of accident can be determined and it can be decided whether or not a crime has been committed.
2. Call ambulance (first bystanders); scene secured (first police); contact relatives of those involved (police).
3. (a) Length of skid marks, final positions of cars, widths of roads and slip lanes, positions of traffic lights and signs, positions of median strips. These would help investigators understand the sequence of events and determine the cause of the accident.
(b) Drivers would be tested for alcohol and any other drugs to see if these may have contributed to the cause.
(c) Cars should have brakes, tyres and lights tested to see if a failure occurred.
(d) "What did you see or hear?"; "Where were you at the time?"; "What were you doing?"; "Do you know either of the drivers?"
Witness reliability may depend on age, driving experience, alertness and any relationship to the people involved.
4. Speed, alcohol, other drugs, tiredness, lack of experience (driver); distractions (driver and passengers); poor road surface (council or road repairer); car defect (driver or car manufacturer); poor visibility, weather conditions (driver or no blame), pedestrian, cyclist (victim).
6. Sudden appearance of low sun from behind a large building of landscape feature, isolated patch of ice, water or oil on road. Warning signs could be put in these places.

2-2 Driving and the law

1. So all road users can use the road safely, with a good idea of what to expect from other drivers.
2. Some description of chaos and dangerous conditions.
3. Punishments vary from state to state.
4. (a) The car is at fault because it is facing a 'Stop' sign.
(b) The van driver should be alert to the other car and driven at speed such that van could stop in time.
5. (a) Because each driver should be alert and drive at a speed to minimise chance of collision.
(b) In rear-end accidents only the second driver has control of keeping a safe distance between the cars.
6. Speed can be calculated from the length of skid marks.

2-3 What is speed?

1. 110 km/h
2. 68 km/h
3. The driver should not be charged because the car was travelling at an average speed of 42 km/h.
4. 10 m/s. This is 36 km/h.
5. 1.7 m/s which is 6 km/h.
6. The family travelled at an average speed of 120 km/h, which is 10 km/h over the limit. If they had travelled at 110 km/h they would have taken 13.6 minutes longer.

2-4 Why should there be a speed limit?

1. Stopping distance increases more than threefold from 19 m to 59 m.
2. Force is increased more than 4 times, like falling from a 3-storey building to falling from a 13-storey building.
3. (a) At 60 km/h stopping distance is 36 m, so car would just be able to stop.
(b) If raining stopping distance would be longer so may not be able to stop.
(c) At 50 km/h stopping distance is 27 m, a much safer margin.
4. Speed limits were reduced to 50 km/h to make roads around houses safer for everyone.
5. Reducing speeds around schools protects young children who do not have the same ability to judge speeds and anticipate dangers.
6. (a) Stopping distances for large heavy trucks is greater because it needs a greater force to stop.
(b) Stopping distances for small cars is less because a smaller force is needed to stop.

2-5 Seat belts

1. The wearing of seat belts should be compulsory because it lowers the number of deaths and serious injuries.
2. The fine encourages the driver to insist all passengers use seat belts.
3. Varies from state to state.
4. Normal use causes abrasion. Ultraviolet radiation from sunlight weakens fibres. To ensure seat

belts are strong enough to hold a person during a crash they must be resistant to these.

5. Seatbelt webbing is tested for strength by placing it under various forces. It should be able to hold a mass of 200 kg, which is larger than most people likely to be able to fit inside a car.
6. Laws exist to protect all people. Family and friends suffer greatly when a loved one is fatally or seriously injured and it costs everyone through taxes.

2-6 Activity: Seat belts and collisions

1. Tests are repeated and averaged to get a more accurate result.
2. As speed increased passengers were thrown forward further, always in the same direction. This is because increasing the speed increases the force with which they move forward.
3. Seat belts.
4. The tape held the passengers in place.
5. Seat belts stop people flying into the windscreen or outside the vehicle during collisions.
6. The passengers keep moving in the original direction because objects will keep moving until a force, such as a seat belt, tape or car interior, acts upon them.

Conclusion: (a) When speed is increased people are in greater danger of sustaining serious injury, as the force of the impact when they stop is also increased.

(b) Wearing seat belts reduces this danger.

2-7 Inertia

1. Headrests prevent whiplash. They should be at the occupant's head height.
2. Whiplash can occur when a stationary car is hit from behind. It can be minimised by having the headrest adjusted correctly to the person's head height and the person's head close to it.
3. Loose objects in the back of a station wagon can fly forward if the car stops suddenly.
4. When a cloth is pulled out quickly from under dishes, the dishes' inertia means that they will remain at rest because there is no force pulling them along.
5. When a car stops suddenly, inertia states that the passenger will keep moving in the same direction until acted on by a force, usually the windscreen, unless the passenger is wearing a seat belt.
6. Driver was probably not wearing a seat belt and collided with the windscreen, causing death.

2-8 Air bags and crumple zones

1. The egg that was protected by the crumpled paper survived the fall; the other did not. This is because the paper was able to slow the egg down gently while the unprotected egg slowed down to rest in a small fraction of a second.
2. (a) eggs = people
(b) crumpled paper = crumple zone
(c) box = car
3. Rubber or plastic bumper bars on cars; polystyrene inserts or particles inside cardboard boxes to protect goods during transport.

4. A 'control' in an experiment is the standard situation against which all other results are compared.

2-9 Reaction distances

1. 83.3 m
2. 25 m
3. 8.3 m longer than 25 m, ie it would take 33.3 m to stop.
4. People under the influence of drugs are less alert so have longer reaction times, so they will travel further in that time.

2-10 Activity: Comparing reaction times

1. Generally reaction time improves with practice.
2. The reaction time is longer if hand is further from ruler.
3. The reaction time is longer if student is distracted.
4. Some people appear to have better reaction times. This may be inherited, due to quicker reflexes or due to medication or health condition.
5. When a person is tired or has taken some drugs, their reaction time may be longer than usual.

Conclusion: Reaction time may be affected by concentration, amount of practice and the distance hands or feet have to move.

2-11 Braking distances

1. (a) Friction can be increased by making surface rougher or increasing pressure between surfaces.
(b) Friction is needed to keep feet on bicycle pedals, to walk, to keep tyres turning on the road and nails in wood.
2. (a) To decrease friction make surfaces smoother, use ball bearings or a lubricant.
(b) Friction is not wanted when ice-skating, in machinery or on door hinges. These all use lubrication to minimise friction. A wheel and axle and gears use ball bearings.
3. (a), (b), (c), (e) and (f) all decrease friction and increase stopping distance; (d) increases friction and so decreases stopping distance.
4. The experiment should use the same vehicle and mass and only vary the conditions as in Q3 above to measure the different stopping distances.
5. Braking time: $100 \text{ km/h} \div 3.6 = 27.8 \text{ m/s}$.
Reducing 8 m/s every sec, $27.8 \div 8 = 3.47 \text{ s}$
Braking distance: average speed = $50 \text{ km/h} = 13.88 \text{ m/s}$ $13.88 \times 3.47 = 48 \text{ m}$
6. Braking time: $90 \text{ km/h} = 25 \text{ m/s}$.
Reducing 5 m/s per sec, $25 \div 5 = 5 \text{ s}$
Braking distance: average speed = $45 \text{ km/h} = 12.5 \text{ m/s}$ $12.5 \times 5 = 62.5 \text{ m}$
7. Braking distance at 120 km/h is 69.4 m ; at 100 km/h is 48.2 m . The difference is 11.2 m .

2-12 A typical traffic accident

1. Accident may have been avoided if Belinda had not been stressed, under the influence of alcohol or distracted by Jason. However, Andy overtook when it was not safe and so would be considered most at fault.

2. Apart from not overtaking, Andy should have been more alert and aware of the road, hill and weather conditions.
3. It would have been better if Belinda had not driven in her emotional state and given herself time to rest, calm down and let her body get rid of some of the alcohol in her blood.
4. Belinda should have made sure Jason was wearing a seat belt. She may have been thinking of other things and not her child. If her son had been conditioned to always wear a seat belt, he may have reminded her of it when he got in the car.
5. Anti-locking brakes release and immediately regrip when a car starts to skid. They may have helped Belinda keep control of the car and avoid skidding into Andy's car.
6. The rain would have reduced visibility and made it difficult for both drivers to see far ahead. Rain can also make the road slippery and a truck will throw up spray into the windscreen of cars immediately behind. Drivers should reduce their speed in rain.
7. The back tyres of Belinda's car must have been lacking sufficient tread, which reduces grip on the road. Having worn tyres increases the chance of skidding.
8. Air bags may have saved Belinda's life, but they wouldn't have prevented Jason's injuries, as he fell sideways out of the car door.
9. Measure the length of all skid marks, width of the road, position of vehicles from top of hill; search car for evidence of drugs, distractions such as drinking vessels or mobile phones. Condition of seat belts, air bags, tyres and lights.
10. Andy should be charged with 'dangerous driving' because he overtook when it wasn't safe to do so.

2-13 Alcohol and driving

1. Blood alcohol (BAC) is a measure of the amount of alcohol in a person's blood. The legal BAC in Australia is 0.05. However, any alcohol will affect a driver to some extent and so it may be said there is no safe level.
2. A standard drink contains about 8-10 grams of alcohol.
3. Some males may be over the legal limit after three standard drinks in the first hour. Some women may be over the limit after two drinks.
4. (a) The only way BAC can be reduced is to wait for the body to break it down.
(b) The liver breaks down alcohol.
(c) About 8 grams of alcohol per hour.
(d) The brain is the first part of the body affected.
5. (a) Both had 3 standard drinks in 3 hours.
(b) Bill was probably the fittest to drive home.
(c) Neither person should be over the legal limit.

2-14 Speed and alcohol revision questions

1. Speed and velocity mean distance covered in a certain time. They are measured in km/h or m/s.
2. (a) 70 km/h
(b) 8 m/s
(c) 28.8 km/h
3. (a) Inertia is the property of objects to remain at rest or motion unless a force acts on the object to change this.
(b) Seat belts, air bags, crumple zones, collapsible steering wheel, anti-locking brakes.
4. (a) Momentum is the measure of the force with which an object moves. It affects force needed to bring an object to rest.
(b) Momentum is calculated as speed $\times$ mass.
5. (a) Total stopping distance = reaction distance + braking distance.
(b) Being tired, under the influence of drugs including alcohol, being distracted, poor visibility, unfamiliar road, poor health.
(c) Poor brakes, worn tyres, wet road, gravel road, heavy load.
6. (a) $75 \div 3.6 \times 1.5 = 31.25$ m
(b) $90 \text{ km/h} \div 3.6 = 25 \text{ m/s}$, reducing speed 15 km/h per second, $90 \div 15 = 6$ s
Average speed = $45 \text{ km/h} = 12.5 \text{ m/s}$;
Braking distance = $12.5 \times 6 = 75$ m
7. (a) Alcohol is a drug because when taken into the body it affects the normal processes.
(b) Side-effects include slow brain function and reflexes, emotional changes - some becoming happy and relaxed, some more aggressive, others depressed and morose. Judgement of speed and distance may be affected, vision can become blurred and reduced to tunnel vision, and balance and general control can be reduced.
8. (a) All Australian states have a legal BAC limit of 0.05 because it is safer than 0.08.
(b) Learner and Provisional drivers have a zero limit. The 0.02 exists to allow for small amounts in some medications.
(c) Learner and Provisional drivers are less experienced and so must remain extremely alert.
9. Try to make them stay the night, call a taxi, or let someone else drive them home.
10. a, d, and e are all equivalent to one standard drink; b, c, and f are two standard drinks.
11. One drink only to be safe.
12. The only way alcohol can be removed from the body is by allowing the liver to break it down.

2-15 Speed and alcohol crossword

1. speed 2. velocity 3. ethanol 4. friction
5. braking distance 6. rough 7. inertia
8. reaction time 9. drinking 10. seatbelt
11. momentum 12. damp 13. skid 14. brakes
15. whiplash 16. liver 17. acceleration
Hidden statement: DON'T DRINK AND DRIVE

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